

Name:

Form:

MYP 1

Year 7

Forces and Motion

Statement of Inquiry:	By exploring the laws of motion and relationships we can describe the movement and interactions of objects.
Key Concept:	Relationships
Related Concepts:	Movement and Interactions



Statement of inquiry
Exploring the laws of motion and relationships which describe the movement and interactions of objects.
Inquiry questions
Factual (what) - What are main types of forces? - What are the effects of a force on an object? Conceptual (how/why) - How are the laws motion used to solve problems? - How can our understanding of the laws of motion be applied in a number of different contexts? - How distance-time graphs are used to describe an objects motion. Contact forces (friction, air and water resistance, upthrust, normal/reaction) and non-contact forces (gravity, magnetism, static electricity). Debatable (is/do) - Is it essential to understand the laws of motion in order to understand the natural world? A force can make an object start or stop moving, speed it up or slow it down, change its direction of motion, or change its shape. Gravity keeps planets in orbit and pulls objects to Earth, friction and water shape rivers and land; Approaches to learning (ATL) magnetism helps some animals navigate. Use Newton's laws. $F = m \times a$ to find force, mass or acceleration, and use balanced/unbalanced forces to explain motion. Information Literacy Skills - Process data and report results. Collaboration Skills - Give and receive meaningful feedback. Yes - forces act in many contexts: sport, transport (braking, air resistance), space and orbits, machinery such as levers, and everyday objects.

7P MYP Command Terms

Term	Definition
Analyse	Break down in order to bring out the essential elements or structure. To identify parts and relationships, and to interpret information to reach conclusions.
Apply	Use knowledge and understanding in response to a given situation or real circumstances
Describe	Give a detailed account or picture of a situation, event, pattern or process
Design	Produce a plan, simulation or model
Discuss	Offer a considered and balanced review that includes a range of arguments, factors or hypotheses. Opinions or conclusions should be presented clearly and supported by appropriate evidence
Document	Credit sources of information used by referencing (or citing), following one recognized referencing system. References should be included in the text and also at the end of the piece of work in a reference list or bibliography
Evaluate	Make an appraisal by weighing up the strengths and limitations
Explain	Give a detailed account
Formulate	Express precisely and systematically the relevant concept(s) or argument(s)
Interpret	Use knowledge and understanding to recognize trends and draw conclusions from given information
Outline	Give a brief account
Present	Offer for display, observation, examination or consideration
Recall	Remember or recognize from prior learning experiences
Select	Choose from a list or group
Solve	Obtain the answer(s) using appropriate methods
State	Give a specific name, value or other brief answer without explanation or calculation
Suggest	Propose a solution, hypothesis or other possible answer
Summarize	Abstract a general theme or major point(s)

7P MYP Glossary Forces and Motion

Forces

Word	Meaning
air resistance	A force that tries to slow things down that are moving through air. It is a type of friction.
contact force	A force that needs to touch an object before it can affect it (e.g. friction).
drag	Air resistance and water resistance are both sometimes called drag.
force	A push or a pull.
force meter	Piece of equipment containing a spring, used to measure forces.
friction	A force that tries to slow things down when two things rub against each other.
gravity	The force of attraction between any two objects. The Earth is very big and so has strong gravity that pulls everything down towards it.
magnetic attraction	A force that attracts objects made out of iron.
magnetic repulsion	A force that pushes two similar magnetic poles apart
Newton (N)	The unit of force.
Newton meter	Another name for a force meter.
non-contact force	A force that can affect something from a distance (e.g. gravity).
speed	How fast something is moving. Often measured in metres per second (m/s).
static electricity	A force which attracts things with extra electrical charges on them.
upthrust	A force that pushes things up when in water.
water resistance	A force that tries to slow things down that are moving through water. It is a type of friction.

Balancing act / Stretching / Weight

Word	Meaning
balanced forces	When two forces are the same strength, but working in opposite directions.
elastic	Any substance that will return to its original shape and size after it has been stretched or squashed.
gram	A unit for measuring mass (g).
kilogram	A unit for measuring mass (kg). There are 1000 g in 1 kg.
mass	The amount of matter that something is made of. Measured in grams (g) and kilograms (kg). Your mass does not change if you go into space or to another planet.
newton	The unit of force (N).
stationary	Not moving.
unbalanced forces	When two forces working in opposite directions are not the same strength.
weight	The amount of force with which gravity pulls something towards the Earth. It is measured in newtons (N).

Friction

Word	Meaning
air resistance	A force that tries to slow things down that are moving through the air. It is a type of friction.
friction	A force that tries to slow things down when two things rub against each other.
lubricant	A substance (normally a liquid) used to reduce friction.
lubrication	Adding a lubricant to something.
water resistance	A force that tries to slow things down that are moving through water. It is a type of friction.

Density, upthrust and floating

Word	Meaning
density	The amount of mass that 1cm^3 of a substance has. Measured in g/cm^3 .
displacement	The volume of water pushed out of the way by an object.
upthrust	A force that pushes thing up when in water.

Speed

Word	Meaning
distance/time graph	A graph that shows how far and how fast something travels during a journey.
kilometres per hour (km/h)	Units for speed when the distance is measured in kilometres and the time is measured in hours.
metres per second (m/s)	Units for speed when the distance is measured in metres and the time is measured in seconds.
miles per hour (mph)	Units for speed when the distance is measured in miles and the time is measured in hours.
speed	How fast something is moving. Often measured in metres per second (m/s).

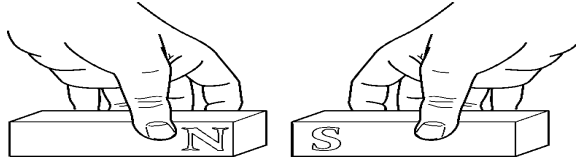
Air Resistance

Word	Meaning
air resistance	A force that tries to slow things down that are moving through the air. It is a type of friction.
streamlined	When an object has a shape that reduces friction and allows it to travel faster.
Accelerate	Gets faster
Decelerate	Gets slower

7P 1a Practical Investigation: Forces in Action

Carry out the following activities and fill in the gaps on this sheet with your results. Some of the words you need are given in the brackets.

1.



The magnets are repelling (attracting/repelling) each other.

The force is strongest when the magnets are close together (close together/far apart).

The name of this force is magnetism (friction/gravity/magnetism).

What will happen if you turn one of the magnets round? They will attract each other instead of repelling.

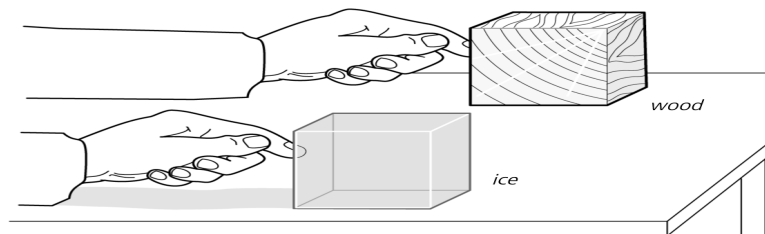
What do 2 North poles do to each other? Two North poles repel each other

What do 2 South poles do to each other? Two South poles repel each other

What does a North pole do to a South pole? Two different poles attract each other

This force acts at a distant and the magnets do not need to touch. It is called a non-contact (contact/non-contact) force.

2.



The force of friction (friction/gravity/magnetism) tries to stop the blocks moving across the desk. It is easier (easier/harder) to push the block of wood, because the wood is smoother (smoother/rougher).

Since the blocks need to touch the desk, this force is called a contact (contact/non-contact) force.

Using the tray, tilt the tray until one of the blocks slides. Which block slides first?

The wooden block slides first because it has less (more/less) friction than the other block.

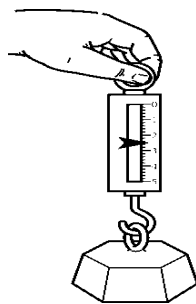
3. Get a plastic rod and some small pieces of tissue paper. Rub the plastic rod with a cloth and hold them close to the tissue paper. Explain or draw what happens in the space below.

The plastic rod attracts (attracts/repels) the small pieces of paper. This type of force is called static electricity. Is this a contact or a non-contact force? non-contact

Extension: Can you explain why this happens?

The rubbed rod becomes electrically charged and attracts (pulls) the small pieces of paper towards it.

4.



Weigh the object using a Newton meter. What is a Newton meter used for?

A Newton meter is used to measure force.

How much does your object weigh? 3 N

What is the mass of the object? 0.3 kg The force of gravity is pulling the object downwards towards the Earth. What is the size of this force that acts on 1 kg? 10 N.

A Newton meter is used to measure force (the pull of gravity on an object), in newtons (N).

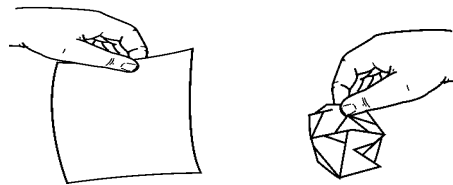
Extension:

The force of gravity is much weaker on the moon. Why?

The Moon has much less mass than the Earth, so its gravitational pull is much weaker.

If you had a mass of 1 kg, find how much it would weigh on the moon?

On the Moon g is about 1.6 N/kg, so a 1 kg mass would weigh about 1.6 N.

5

The paper takes less time (less time/more time) to fall if it is crumpled up.

The force trying to slow down the paper is called air resistance. Is this a contact or a non-contact force? It is a contact force because the paper is falling and pushing through the air.

Extension: This force is slowing down the paper. Which bit of paper is experiencing more of this force and why? The flat sheet - it has a bigger surface area, so it hits more air particles and gets more air resistance.

6 Get a plastic bowl and half fill it with water. Put a ping pong ball and a marble in the water. Draw a diagram or explain in the space below, what happens.

The ping pong ball floats (upthrust = its weight); the marble sinks (its weight is greater than the upthrust).

The force of the water pushing upwards is called upthrust, if this force is equal or larger than the force due to gravity then the object in the water will float.

Extension: Explain what water resistance is? Does this get greater or smaller if a speed boat goes faster? Water resistance is a force (a type of friction) that slows things moving through water. It gets greater as the speed boat goes faster.

Check your understanding:

Consider the six practical's you have just completed and tick the correct box to show the level of your understanding so far – be honest!

I can.....	✓ Expert	Good	Reasonable	Novice
Describe what forces do.				
Recall the names of contact and non-contact forces.				
Explain how to measure forces.				
Explain the difference between mass and weight				

7P 1b Examples of Forces

Card Matching Activity:

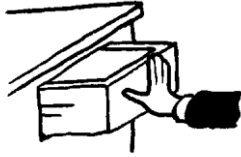
You will be given some cards. You must match up the name of force, the description and the photographs of the examples.

Record the information in the table below:

Force	Description	Examples
Gravitational	The force of attraction between any two objects; the Earth's gravity pulls everything down towards it.	An apple falling from a tree; a ball thrown up falling back down.
Magnetic	A force that attracts objects made out of iron (magnetic materials).	A magnet picking up paper clips; a fridge magnet holding a note on the door.
Friction	A force that tries to slow things down when two things rub against each other.	Rubbing your hands together to warm them; bike brakes slowing a wheel.
Upthrust	A force that pushes things up when they are in water.	A rubber duck floating on water; a boat floating on the sea.
Air Resistance	A force that tries to slow things down that are moving through air; it is a type of friction.	A parachute slowing a skydiver; a flag flapping in the wind.
Static Electricity	A force which attracts things that have extra electrical charges on them.	A rubbed balloon sticking to a wall; a rubbed rod picking up small pieces of paper.

7P 1c: Types of Forces (Lesson 1 Plenary)

- 1 Which forces are pushes and which ones are pulls? Write the correct words in the spaces.



push



pull



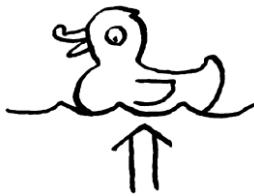
push



pull

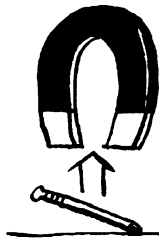
- 2 Look at the pictures below. Write the names of the forces next to the arrows.

A



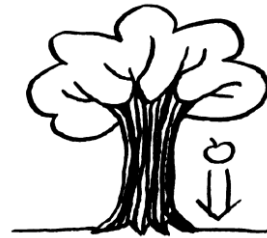
gravity

B



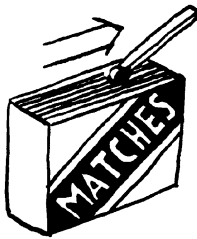
magnetism

C



static electricity

D



upthrust

E



friction

F



air resistance

- 3 Which pictures above are examples of non-contact forces? A, B and C

Which pictures above are examples of contact forces? D, E and F

Home-work Assignment:

Extension: What happens to an object when 2 forces are balanced? Draw a diagram below of a car and use arrows to represent the forces and explain what is happening to the car. *The car stays still, or carries on moving at a steady speed in a straight line - balanced forces do not change motion.*

7P 2a Investigating Weight using a Newton Meter

Aim:

You will experimentally investigate the relationship between the weight of an object and its mass.

Method:

1. Set up a Newton-Meter (also called a Force-Meter) as per teacher guidance using a clamp stand, and G-clamp.
2. Using slotted masses you choose 6 different masses to hang, and using the Newton Force Meter measure the corresponding force of weight exerted by each mass in Newton's. Each slotted mass has a mass of 0.1kg.
3. Construct an appropriate table to record your results below.
4. Add the name and units to the headings.
5. To make your data more **reliable**, measure the weight twice and take an average.

Table of Results:

Mass added to Newton-Meter			
/ Kilograms			
0 kg	0 N	0 N	0 N
0.1 kg	1 N	1 N	1 N
0.2 kg	2 N	2 N	2 N
0.3 kg	3 N	3 N	3 N
0.4 kg	4 N	4 N	4 N
0.5 kg	5 N	5 N	5 N
0.6 kg	6 N	6 N	6 N

Analysis of Results and Conclusion:

1. **Construct a graph** to show the relationship between the force of weight and mass. This should be done on graph paper and the following *criteria addressed*;
 - *Title of graph included*
 - *Label the x and y axis with correct units*
 - *Scales should be appropriate*
 - *Points plots accurately and connect using curve/straight line of best fit*
- Plot mass (kg) on the x-axis and weight (N) on the y-axis; draw a straight line of best fit through the origin.

2. How are mass and weight related?

Mass is the amount of matter an object is made of, measured in kilograms (kg)

Weight is the force of gravity pulling on that mass, measured in newtons (N)

Weight = mass $\times$ 10, so weight is directly proportional to mass.

3. What happens when you increase the mass of an object in relation to the weight of the object? Support your answer with data from the inquiry.

When the mass increases, the weight of the object increases in proportion.

For example 0.1 kg had a weight of 1 N and 0.4 kg had a weight of 4 N

So doubling the mass doubles the weight - the graph is a straight line through the origin.

Reflection/Evaluation:

Write a short paragraph below describing the experiment, what you learned from the experiment and how it affects your everyday life. What does a well-organized scientist always do when they take measurements?

We hung different slotted masses on a Newton meter and measured the weight of each mass twice.

I learned that weight is proportional to mass: 1 kg weighs about 10 N on Earth

This affects everyday life because shops sell food by mass and we feel its weight.

A well-organised scientist always repeats measurements and takes an average.

They also write down the units and check that the results are reliable

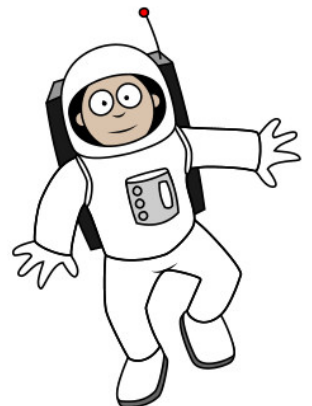
Extension

An astronaut has a mass of 70kg and a weight of 700N here on Earth.

The astronaut flies far away from the Earth.

The mass of the astronaut is now greater / less / the same but the weight of the astronaut far from Earth is greater / less / the same.

In space we say the astronaut is weightless.



7P 2b Lines of Best Fit – Extra Help

Aim:

Choosing the most appropriate Line of Best Fit.

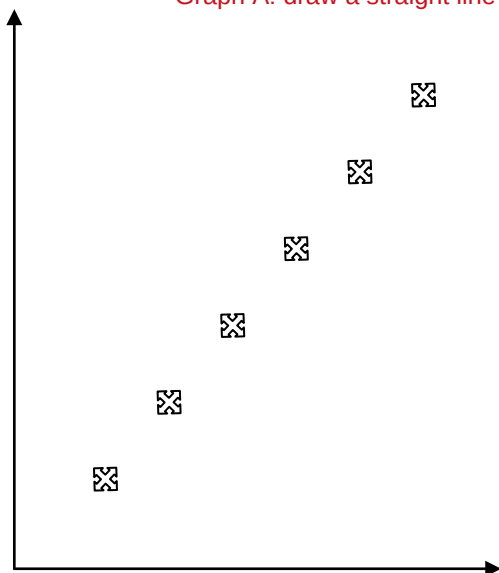
Method:

Draw the Line of Best fit for each graph. Use a ruler where needed.

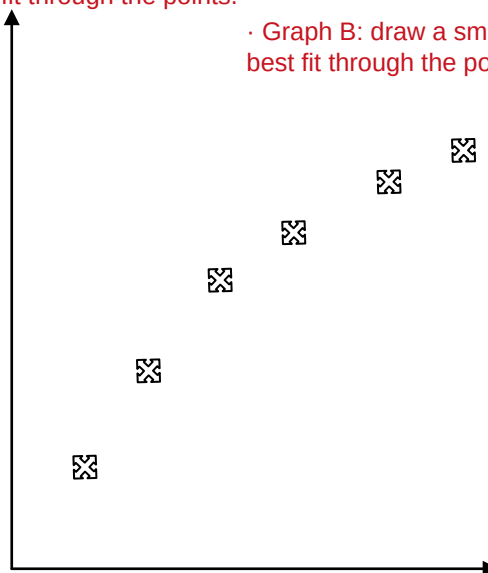
Check your graphs with your teacher.

Graph A

· Graph A: draw a straight line of best fit through the points.

**Graph B**

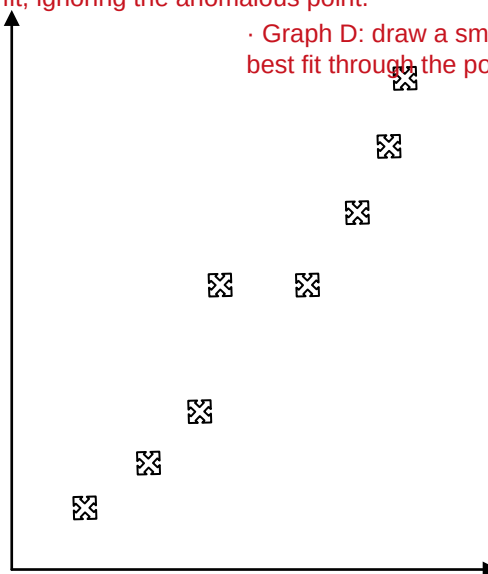
· Graph B: draw a smooth curve of best fit through the points.

**Graph C**

· Graph C: draw a straight line of best fit, ignoring the anomalous point.

**Graph D**

· Graph D: draw a smooth curve of best fit through the points.

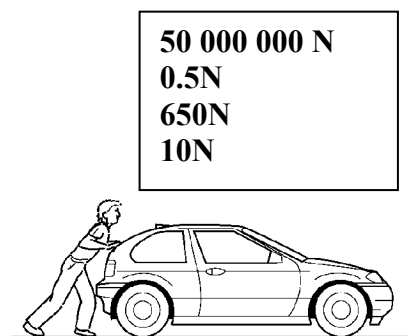
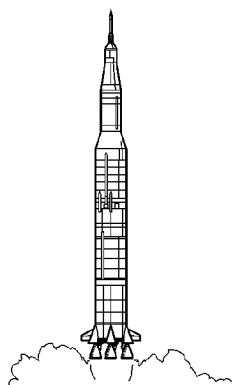
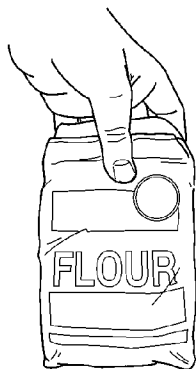


7P 2c Weight, Gravity and Mass - Review

1a The unit of force is called the newton.

1b The unit is given a capital letter because it is named after a scientist (Isaac Newton)

2 The pictures below show examples of forces. Write the likely size of each force in the space below each picture. Choose numbers from the box.



50 000 000 N
0.5N
650N
10N

0.5 N

650 N

10 N

50 000 000 N

3 Fill in the missing words:

The Earth pulls all objects towards it. This is because of

gravity. It is this effect which gives us our weight.

4 Complete these sentences using words from the box:

The weight of a 2 kg mass on Earth is 20 N.

The typical mass of a man on Earth is about 80 kg.

The unit of mass is the kilogram.

The unit of weight is the Newton.

The mass of a 2 kg bag of apples on the Moon is 2 kg.

The weight of a 2kg bag of apples on the moon is 3.2 N.

kilogram	Newton	20 N
2 kg	80 kg	3.2 N

- 5 Explain in your own words, the difference between **mass** and **weight**.

Mass is how much matter an object is made of, measured in kilograms (kg).

Weight is the force of gravity pulling on that matter, measured in newtons (N).

Mass never changes, but weight changes if the strength of gravity changes.

- 6 If an astronaut leaves the Earth and flies into space describe what happens to their **mass** and **weight**. How does an astronaut feel when they are far from the Earth?

Their mass stays the same because the amount of matter has not changed.

Their weight gets smaller because gravity is weaker far from the Earth.

The astronaut feels weightless and floats around inside the spacecraft.

Extension: Find out about the force due to gravity on other planets or the moon. How is the gravitational force related to the mass of the planet?

The greater the mass of the planet, the stronger its gravitational pull:
the Moon pulls with about 1.6 N/kg and Earth with 10 N/kg.

7P 2d Balanced and Unbalanced Forces

Use these words to fill the gaps.

water

opposite

float

upthrust

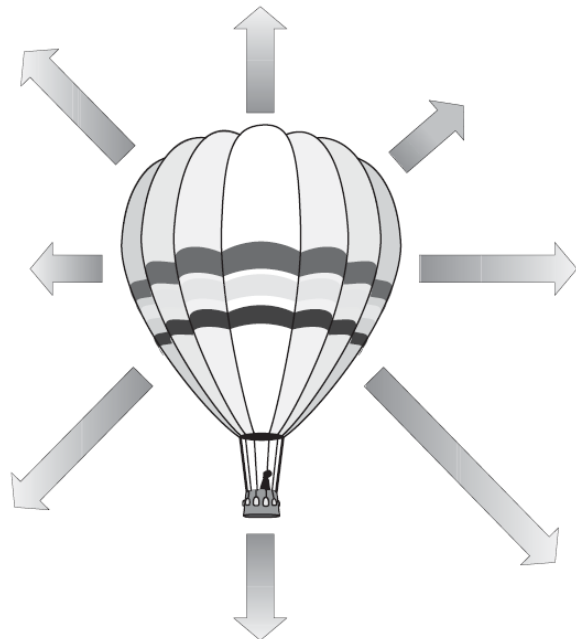
balanced

- a** If the forces are _____ an object will not move. *balanced*
- b** Balanced forces are the same size and pull in _____ directions. *opposite*
- c** When you put an object in water, the _____ pushes upwards on the object. This push is called _____.
water
upthrust
- d** If the upthrust on an object is the same size as its weight, the object will _____.
float

- 2** This balloon is floating in the air. Look carefully at the diagram.

Which arrow shows the upthrust?

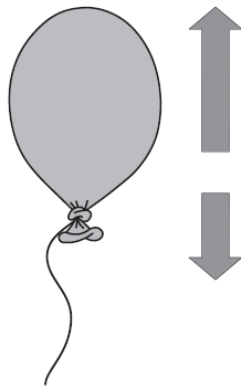
Circle the arrow.



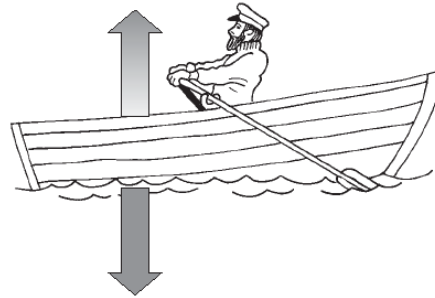
· Circle the arrow pointing straight upwards on the balloon - that is the upthrust.

- 3 Look at these diagrams. The arrows show the size and direction of the forces.

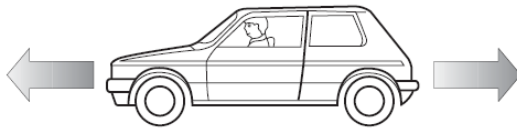
A



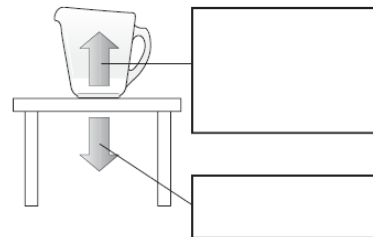
B



C



D



- a In which diagrams are the forces balanced? _____

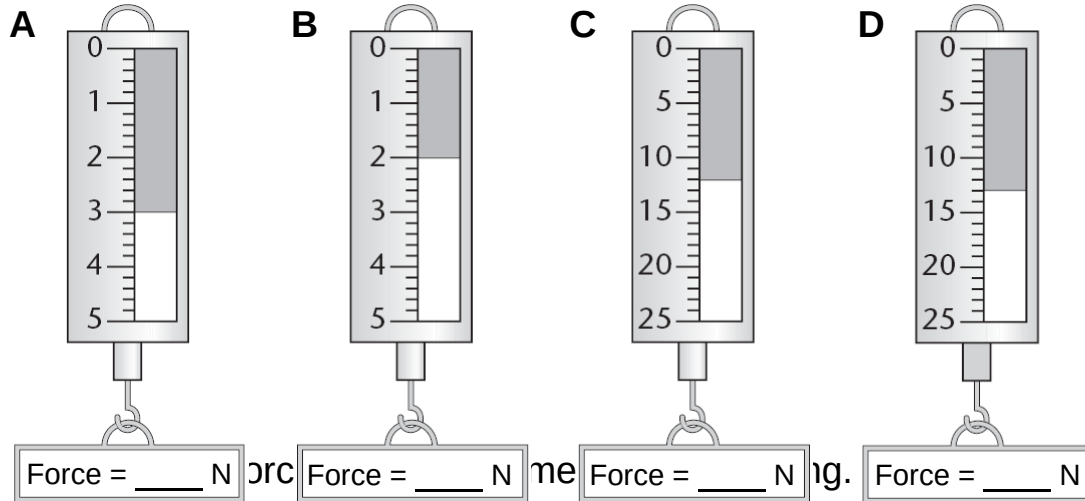
Diagrams A and D - the arrows are the same size, so the forces are balanced.

- b Will the boat float or sink? The boat will float, because the upthrust is the same size as its weight.

- c The jug is standing on a table. The *weight* of the jug is the same size as the *reaction force* of the table. Label the forces in diagram D. Label the downward arrow 'weight' and the upward arrow 'reaction force of the table' (or 'upthrust').

7P 2e Weight and Gravity

- 1** Look at these Force-Meters (Newton-Meters). Each is measuring a different force. Forces are measured in Newtons (N).



- b** Which of these forces is the biggest? _____ N
c Which of these forces is the smallest? _____ N
- 2** Match the words to the descriptions.

The biggest force is 6 N (Force-meter C).

force

How much material **mass**
something is made of.

The smallest force is 2 N (Force-meter A).

weight

The pull of the Earth's **weight** gravity on something.

mass

A push or a pull. **force**

gravity

Weight is measured in these. ~~Newtons (N)~~

Newtons (N)

The units of mass. kilograms (kg)

kilograms (kg)

The effect that keeps the earth gravity orbiting the sun.

2 N

4 N

6 N

3 N

- 3 Look at this table. It contains information about the weight and mass of some items.

Item	Mass in kg	Weight in N
James	47	470
large book	1	10
bag of sugar	2	20
bag of potatoes	10	100
Janine	53	530

Write *true* or *false* for each sentence.

- a Mass is measured in kilograms. True
- b Weight is measured in kilograms. False - weight is measured in newtons.
- c Weight is caused by gravity pulling down on a mass. True
- d 1 kilogram has a weight of Newtons. True - 1 kg has a weight of 10 N.

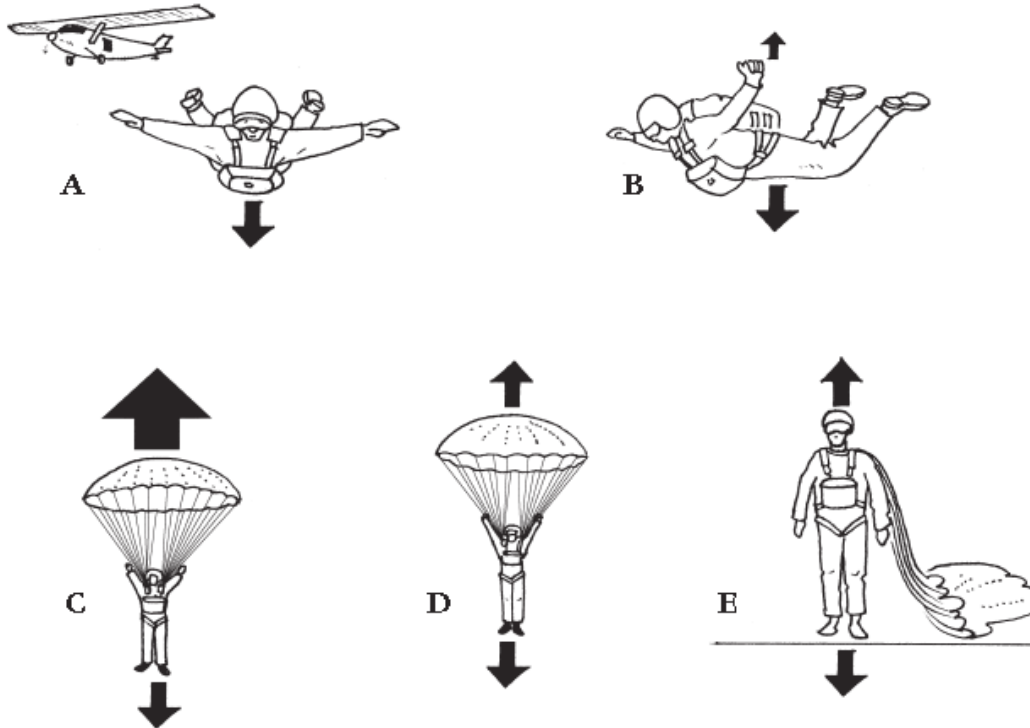
Look at the table.

- e Which item has a mass of 10 kg? bag of potatoes
- f Which item has a weight of 100 N? bag of potatoes
- g Which item has the smallest mass? large book (1 kg)
- h Which item has the smallest weight? large book (10 N)
- i Which item has the largest weight? Janine (530 N)
- j Which item has the largest mass? Janine (53 kg)
- k James has a smaller / larger mass than Janine.
- l Janine weighs more than / less than James.

Underline the right words.
Cross out the wrong words.

7P 2f Parachute Jump

These pictures show different stages in a parachute jump.



- 1 What is causing the downwards forces shown in the pictures?
- 2 What is causing the upwards force in pictures B to D?
- 3 What is causing the upwards force in picture E?
- 4 In which pictures is the skydiver gaining speed?
- 5
 - a Which picture shows that her air resistance has suddenly increased?
 - b What has caused the sudden increase in her air resistance?
 - c What will happen to her speed?
- 6 Which two pictures show balanced forces?

· Parachute jump diagram: label the upward arrow as air resistance and the downward arrow as weight on the skydiver.

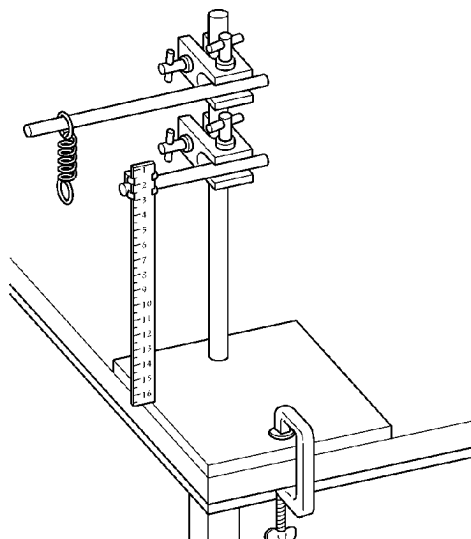
7P 3a Investigating Stretching

Apparatus:

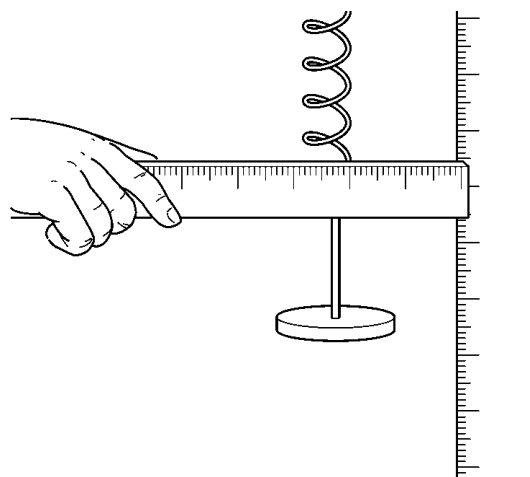
- Stand and 2 clamps
- Weight holder and 1N weights
- Metre rule
- G clamp
- Spring
- Objects A, B and C with unknown masses

Method:

1. Set up your apparatus like this:



2. Measure the position of the bottom of the spring. You can use another ruler to help you. Write your result in the table.
3. Hang the holder on the spring and measure the length again. The holder has a weight of 1N. Write your result in the table.
4. Add another weight on the spring and measure its length. Continue up to 5 weights.

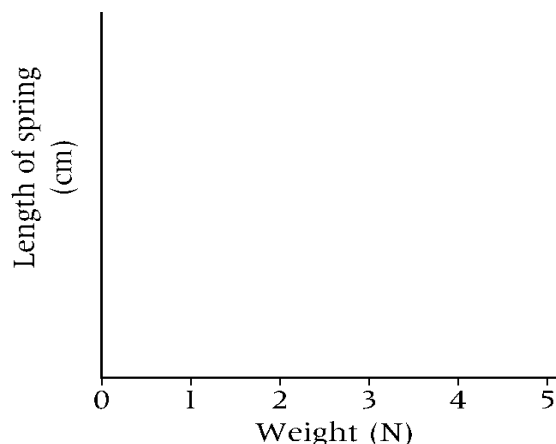


Results:

Weight (N)	Length of spring repeat 1 (cm)	Length of spring repeat 2 (cm)	Average length of spring (cm)
0	10.0 cm	10.0 cm	10.0 cm
1	12.0 cm	12.2 cm	12.1 cm
2	14.0 cm	14.2 cm	14.1 cm
3	16.1 cm	16.1 cm	16.1 cm
4	18.0 cm	18.2 cm	18.1 cm
5	20.1 cm	20.1 cm	20.1 cm

Graph:

Plot a graph of your results on graph paper. You will need to use axes like these:



· Plot weight (N) on the x-axis and length of spring (cm) on the y-axis, then draw a straight line of best fit.

Conclusion:

What happens to the length of the spring when you increase the weight? (Extension: What about when you double the weight?)

As the weight increases, the length of the spring increases

The extension is directly proportional to the weight - the graph is a straight line.

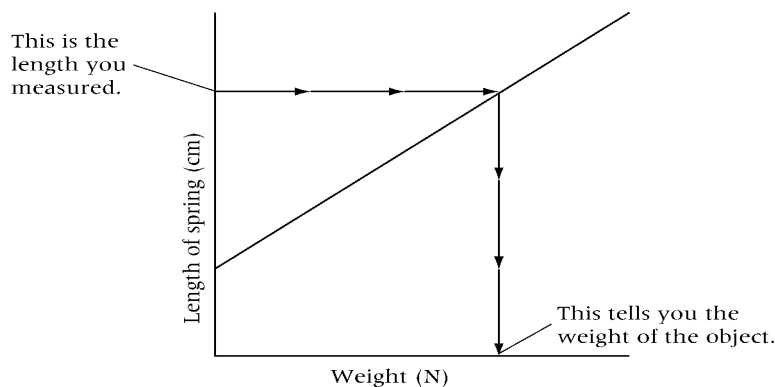
For example 1 N stretches the spring by 2 cm and 2 N stretches it by 4 cm.

So doubling the weight doubles the extension of the spring

This is because the spring obeys Hooke's Law (up to its elastic limit)

Using your Graph to find Unknown Weights:

Now you can use your force meter to weigh other things. There are 3 objects to measure A, B and C. Hang an object on the spring and measure its length. Then use your graph to find its weight, like this:



Unknown Object	Length (cm)	Weight (N)	Actual Weight measured using a Newton-meter (N)
A	14.0 cm	2.0 N	2.0 N
B	17.0 cm	3.5 N	3.5 N
C	19.0 cm	4.5 N	4.5 N

Conclusion:

Did this technique get a reasonably accurate value for the weight of the unknown objects?

Do we use springs to measure weights?

Yes - a Newton meter contains a spring, and the extension of the spring shows the size of the weight.

Extension Activities:

Sketch a diagram and label the forces on the mass using arrows. Are they balanced or unbalanced? (Hint: Label one of them 'Pull of the spring').

Find out about Hooke's Law. Does it apply to all materials that can be stretched? Does this experiment obey Hooke's Law?

The mass has the pull of the spring upwards and its weight downwards; they are equal, so the forces are balanced.

Hooke's Law: extension is proportional to force. It only applies to elastic materials up to the elastic limit.

Yes, this experiment obeys it (straight line graph).

7P 4a Which Shoes Have the Most Friction?

Predict:

What type of shoes has the most grip or friction? Is it the shape or material of the shoe? Predict and explain whose shoe in your group you think would have most friction and why?

I predict the shoe with the deepest tread and softest rubber sole (the trainer) will have the most friction.

Because a rougher, softer sole grips the surface better, so a bigger force is needed to pull it.

I predict the smooth-soled school shoe will have the least friction.

Apparatus:

- 2 or 3 different types of shoes (if someone in your group has a training shoe it would be good to compare)
- Force meter and string.

Method:

1. Attach the Newton meter to the first shoe (use the laces or attach using a loop of string around the base of the shoe).
2. Choose a surface that you are going to pull the shoes along. You could use the bench or the floor, but you need to use the same surface for each test.
3. Gently pull the shoe along the surface. Try to pull it at a constant speed. Read the force meter, and write down the force needed to pull it.
4. Repeat your measurements 2 more times, and write the results in the table. Calculate the mean (average) force needed.
5. Repeat each step for other types of shoe.

Results:

Shoe description	Force needed to pull shoe (N)			Mean force (N)
	1st pull	2nd pull	3rd pull	
Trainer with deep tread	4.5 N	4.6 N	4.4 N	4.5 N
Smooth-soled school shoe	2.0 N	2.1 N	1.9 N	2.0 N
Football boot with studs	5.8 N	5.9 N	5.7 N	5.8 N

Analysing Your Results:

Find the mean (average) by adding up all three results and then divide your answer by 3. Show your results in a **bar chart** and ensure that you address the following criteria:

- *Title of graph included*
- *Label on x and y axis with correct units*
- *Scales should be appropriate*
- *Points plots accurately and connect using curve/straight line of best fit*

Conclusions

1. Which shoe had the most friction? Can you explain why?

The football boot (with studs) had the most friction - it needed the largest force to pull it.

Its sole is the roughest and grippiest, so more surfaces catch on the floor.

More friction means a bigger force is needed to make it slide.

2. Which shoe had the least friction? Can you explain why?

The smooth-soled school shoe had the least friction - it needed the smallest force.

Its sole is the smoothest, so few surfaces catch on each other.

Less friction means less force is needed to pull the shoe along.

Evaluation

Please justify and explain the points you make. Here are a few points to consider:

- Was this a fair test?
- Was it reliable?
- Were there any problems carrying out the experiment?
- How could this practical be improved?

It was a fair test because we used the same surface, the same shoe type and the same Newton meter for each pull.

We repeated each pull three times and took a mean, which makes the results more reliable.

A problem was keeping the shoe moving at a constant speed while reading the force meter.

The shoe sometimes juddered, so it was hard to read the force accurately.

Different people pulled at slightly different angles and speeds.

We could improve it by using a motor or a data logger to pull the shoe at a steady speed.

We could also test each shoe on more surfaces, such as carpet, wood and tiles.

Taking more repeats and having the same person pull would reduce errors.

The repeats were close together, so the results are reliable and agree with the prediction.

The prediction was supported: the shoe with the deepest tread needed the largest force.

Overall the mean force gives a good way of comparing friction between the shoes.

Mostly fair - the same surface and Newton meter were used, but the shoes differ in mass and sole size, so mass should be kept the same.

Reasonably reliable - each shoe was tested 3 times and averaged; more repeats and readings taken the same way would improve reliability.

The shoe may not move at a constant speed and the meter may be read at an angle; pull smoothly and repeat, keeping the surface the same.

Friction Summary Questions

1. Complete the following sentence:

Friction is a force which opposes motion.

2. Outline what causes friction. You may wish to include a diagram.

Friction is caused by the tiny bumps and rough parts on two surfaces catching on each other.

When the surfaces rub together these bumps grip and resist the movement

The rougher the two surfaces, the bigger the friction force between them.

A force is needed to overcome this grip and keep the object moving

3. Give some examples of when it is useful to have lots of friction and when it is a nuisance. Add examples to the table below.

Useful Friction	Nuisance Friction
Gripping the ground when you walk	Wear on the soles of your shoes
Bicycle and car brakes slowing the wheels	Machinery parts getting hot and wearing away
Rubbing a match to light a candle	Rubbing your hands together too hard causing blisters or burns
Tyres gripping a wet road	Energy wasted pushing a ship or plane through air or water
Holding a pencil so it does not slip when you write	Squeaky door hinges that need oil

Extension Questions

Draw a diagram of your shoe and label the forces acting on it, call the forward force 'pull'. (don't forget the up and down forces as well as the forward and backward ones)

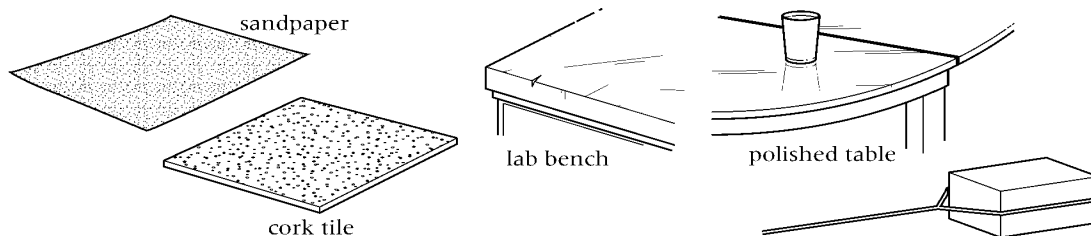
Shoe diagram: label 'pull' forwards, 'friction' backwards, 'weight' downwards and 'reaction force of the ground' upwards.

If you were designing a new sports shoe, what type of material would you use and what type of grip would you design?

I would use a soft, rubbery, flexible material and design a deep tread/studded grip so it grips the ground and does not slip.

7P 4b Friction around Us

1 Look at these pictures.



You pull the block along all of these surfaces.

It will be easiest to pull the block along the smooth polished surface (e.g. the glass/table top)

This is because it has the smoothest (smoothest/roughest) surface, so there will be the least (most/least) friction.

It will be hardest to pull the block along the rough surface (e.g. the carpet)

This is because it has the roughest (smoothest/roughest) surface, so there will be the most (most/least) friction.

2. Complete these sentences and fill in the gaps.

Friction is a force caused by two surfaces moving over each other.
Rough surfaces produce more friction than smooth surfaces.

Friction can be helpful, because it stops us slipping when we walk, and brakes that slow down cars or bicycles could not work without friction.

Friction is not always useful. It can cause heat and noise, and wear away things that rub against each other.

Oil can reduce the amount of friction. Oil stops the surfaces touching each other.

Liquids that reduce friction are called lubricants.

Extension Activities:

Find out why Formula 1 teams need to change the tires on the cars depending on if the race track is wet or dry.

What has this got to do with friction and the speed of the cars? Why is it dangerous to use dry weather tires when the race track is wet?

In wet weather F1 teams use tyres with grooves that push the water away, so the car still grips the track.

Dry-weather (slick) tyres have no grooves, so on a wet track they lose grip and the car can skid out of control.

7P 5a Static Electricity

Introduction:

Static charge can build up when two surfaces rub against each other. A simple experiment can show this effect. When you rub a balloon on your jumper, electrons move from the jumper onto the balloon. These extra electrons make the balloon negatively charged. The jumper is now positively charged. The static charge on the balloon can allow you to stick it to a wall!



Questions:

1. When does static charge build up?

Static charge builds up when two surfaces rub against each other.

Electrons are transferred from one surface to the other, giving each a charge.

2. What are the two types of electric charge?

Positive charge

Negative charge

3. What particles move to transfer static charge?

Electrons (tiny negative particles) move.

They are transferred from one material to the other when they are rubbed together.

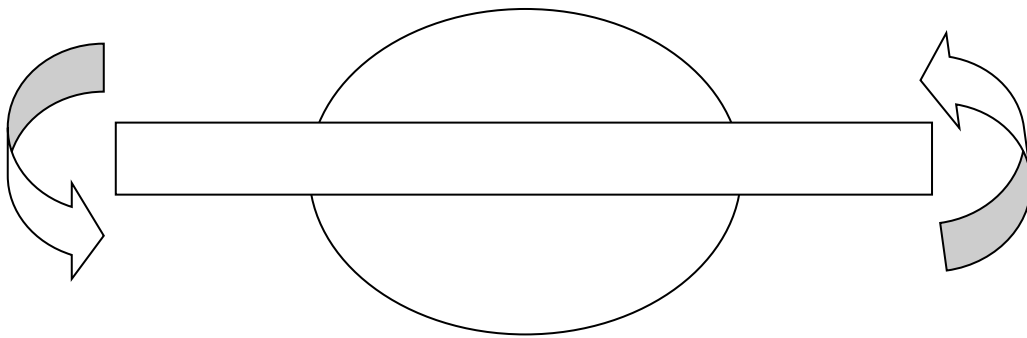
7P 5b Investigating Static Electricity

Aim:

To investigate the behaviour of electrically charged polythene and acetate strips.

Method:

1. Charge the white polythene strip by rubbing with a cloth.
2. Hold the strip close to some small pieces of paper to check that it has a static charge.
3. Now place the charged strip on a watchglass.
4. Charge another white strip and bring it up to the end of the strip on the watchglass. Watch what happens.



5. Next, charge an acetate strip and bring it close to the white strip on the watchglass. Again, watch what happens.

The polythene and acetate strips have opposite charges on them.

- Describe the experiment with the charged strips and watchglass under the following headings: Diagram, Method, Results and Conclusion. Your teacher may give you some guidance.
- What do you think would happen if you repeated the experiment with two acetate strips? If time allows, try this out!

The electric ray possesses two large electric organs on each side of its head!
It can deliver over 300 volts

Diagram: draw the polythene strip balanced on the watchglass with the second charged strip above it.

Method: rub both strips with a cloth to charge them, then bring the second strip close to the first and watch.

Results/Conclusion: two polythene strips have the same charge so they repel each other; a polythene and an acetate strip have opposite charges so they attract.

Two acetate strips would also have the same charge on them, so they would repel (push away from) each other.

Diagram:

- Draw a labelled diagram of the apparatus used in the investigation (e.g. the object, the newton meter and the bench).

Method:

1. Set up the apparatus as shown in the diagram, ready to measure the force.
2. Attach the object to a newton meter.
3. Pull the object at a steady speed and read the force shown on the newton meter.
4. Record the reading in the results table.
5. Repeat the measurement three times and calculate the average force.
6. Change one factor (e.g. the mass of the object) and repeat steps 2-5.
7. Keep all other factors the same so the test is a fair test.

Results:

- Draw a results table with headings (e.g. Mass added, Force in N), showing repeats and the average force.

Conclusion:

- The results show that as the mass of the object increased, the force needed to move it also increased.
- This is because a larger mass has a larger weight (force of gravity), so a bigger force is needed.
- The graph of the results is a straight line through the origin, showing the variables are directly proportional.
- The investigation was a fair test because we kept the equipment and the method the same throughout.
- Repeating each measurement and taking an average improved the reliability of the results.

7P 5c Van der Graaf Generator Demonstration

Your teacher will demonstrate the Van der Graaf generator in action.



Make some notes describing the demonstrations. Try to include an explanation of the Science behind each. There is space at the bottom of the page to include some diagrams and sketches.

Draw sketches of the demonstrations, e.g. the generator dome with sparks and a student's hair standing on end inside it.

2. When a student touches the dome, their hair stands on end because the charge spreads through the hair.

Each hair now has the same type of electric charge on it.

Like charges repel, so the hairs push each other apart and stand up.

3. Sparks can be seen and heard when a hand is held near the dome, because the charge jumps across the gap.

The air becomes able to conduct electricity, letting the charge flow across as a spark.

4. The charge on the dome is made by electrons being transferred onto it as the belt rubs against it.

This shows static electricity, which is a non-contact force (the electrostatic force).

7P 6a Practical Investigation: Magnetism

Part 1: Magnetic Materials

- 1 Test the materials using the magnets provided and fill your results into the following table.

wood	iron	plastic	aluminium paper	copper	steel	nickel
------	------	---------	--------------------	--------	-------	--------

Magnetic materials	Non-magnetic materials
iron	wood
steel	plastic
nickel	aluminium
	copper
	paper

- 2 a What does 'attract' mean when with reference to magnets?

Attract means to pull towards - a magnet pulls magnetic materials towards it.

- b What does repel mean?

Repel means to push away - two like poles of magnets push each other apart.

- c Do magnetic materials attract or repel a magnet? How can you get a magnet to repel an object?

Magnetic materials are attracted to a magnet - they are pulled towards it.

To repel an object you need another magnet: put two like poles (north-north or south-south) together.

- 3 Write true or false next to each of these sentences.

a Magnets only attract magnetic materials. True

b Magnets can attract or repel other magnets. True

c Magnets repel non-magnetic materials. False

d Magnetism goes through paper. True

e Magnetism goes through iron. True

f The two ends of a magnet are called the east and the west poles. False

- 4 a If you put two north poles together, will they attract or repel?
- b If you put two south poles together, will they attract or repel?
- c If you put a north and a south pole together, will they attract or repel?

They will repel each other (two north poles are like poles).

Extension: Research and find out if there are any other materials that are magnetic. Why are only these certain materials magnetic?

Part 2: Making a Magnet

Method:

1. Straighten a paper clip and stroke it, in one direction, about 20 times with one pole of a bar magnet.
2. Label one end of the paperclip A and the other end B.
3. Straighten out a second paper clip and repeat step 1 and 2..
4. Pick up one of the paper clips and leave the other on the bench.
5. Touch an end A of one paper clip to the other end A of the other paper clip. Then try an end B touching another end B. Finally touch an end A to an end B.
6. Record your results in the table below.

End of paper clip	Force of attraction?
Repel (no attraction) A + A	
Repel (no attraction) B + B	
Attract A + B	

They will repel each other (two south poles are like poles).

Conclusion:

1. Which ends attracted each other?

End A and end B attracted each other (the opposite ends of the paper clips)

They will attract each other (north and south are opposite poles).

2. Why do you think the other ends didn't repel?

The paper clips are only weakly magnetised, so the repulsion between the like ends was too small to be seen.

When the same ends were brought close together they attracted each other instead

A magnetised object attracts other magnetic objects, which hid the weak repulsion

Other magnetic materials include cobalt and alloys containing iron, nickel or cobalt (e.g. alnico).

3. Are the paper clips acting as magnets? How do you know?

Yes - the paper clips are acting as magnets. We know because they attract and repel other magnetised paper clips.

This shows they have a north pole and a south pole, just like a bar magnet.

Only materials containing iron, nickel or cobalt are magnetic, because their atoms can line up to form magnetic domains.

4. Can you think of a way to make the paper clips into stronger magnets?

Stroke the paper clips many more times with the bar magnet (e.g. 50-100 strokes), always in the same direction.

Using a stronger bar magnet would also make stronger magnets

More stroking lines up more of the magnetic domains, so the magnetism becomes stronger

5. Using diagrams, explain what is meant by a magnetic domain?

· Draw two diagrams: one showing magnetic domains pointing in random directions (unmagnetised) and one showing them lined up in the same direction (magnetised).

A magnetic domain is a region where the tiny atomic magnets in the material all point in the same direction.

6. Can magnetism be removed from a magnetised object? And if so, how?

Yes - magnetism can be removed by heating the object, dropping or hitting it, or stroking it with a magnet in the opposite direction.

These actions knock the magnetic domains back into random directions

Once the domains no longer line up, the object loses its magnetism

Part 3: Magnetic Field Lines

A magnetic field is the space around a magnet where it has an effect. The magnetic field of a bar magnet will make compass needles point in different directions. Magnetic field lines can be identified using a plotting compass, or iron filings as your teacher will demonstrate to you. If time permits you can complete this practical in student groups.

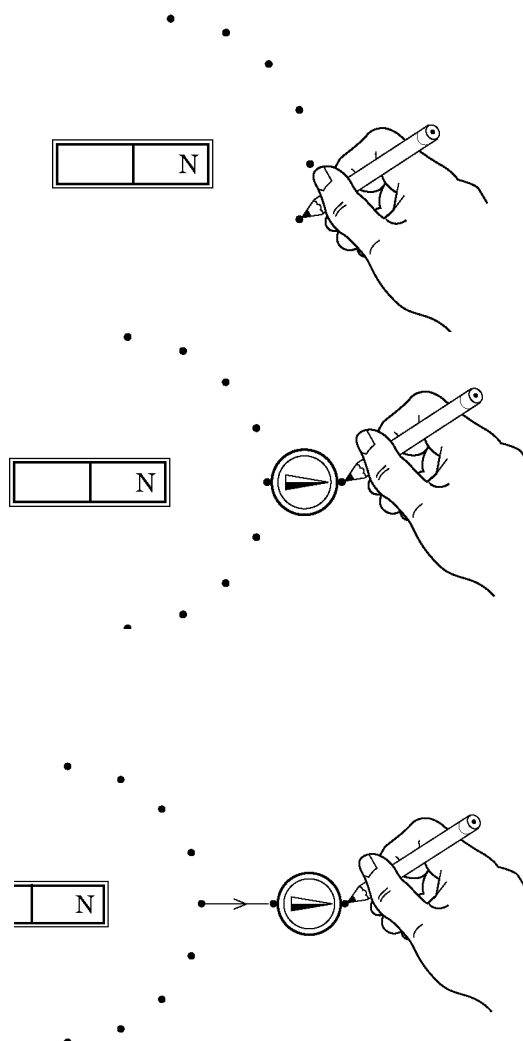
• Complete the diagrams by drawing the magnetic field lines around the bar magnet as smooth curves with arrows pointing from the north pole to the south pole.

Apparatus:

- Bar magnet
- Plotting compass/iron filings
- Sheet of paper
- Pencil

Method:

- 1 Put the magnet in the middle of the piece of paper and draw around it. Draw nine dots around the north end of the magnet, like this:
- 2 Do not move the magnet while you are doing this practical work. Put a plotting compass on the paper so that its tail is over one of the dots. Draw a dot on the paper near to the point of the compass.
- 3 Join the two dots and mark the line with an arrow. Then move the compass so that its tail is over your new dot. Make another dot next to the point.
- 4 Repeat step 3 until you come to the edge of the paper.
- 5 Now repeat steps 2 to 4 for the other dots that you drew around one end of the magnet.



The arrows on your lines show which way a north pole would move. They point away from the north pole of the magnet. The magnetic field goes from north to south.

Extension: Conduct independent research and discuss applications of magnetism – for example in electric bells and replay switches.

Check your understanding:

Consider the three practical's you have just completed and tick the correct box to show the level of your understanding so far – be honest!

I can.....	Expert	Good	Reasonable	Novice
Describe the force of magnetism				
Recall examples of magnetic materials				
Explain what a magnetic domain is				
Explain what magnetic field lines represent.				

Are there any further questions you have on magnetism that you would like answered?

e.g. Why does cutting a magnet in half give two magnets? Why do opposite poles attract? How does Earth's magnetic field protect us from the solar wind?

7P 7a Floating, Sinking and Density

Research Question:

Is there a connection between the density of an object and whether it floats or sinks?

Density measures how much mass there is in 1 cm^3 of something. You have to work it out from a mass and a volume.

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

Yes - an object floats if its density is less than the liquid's density (1 g/cm^3 for water) and sinks if its density is greater than 1 g/cm^3 .

E.g. An object has a mass of 36g. It has a length of 4cm, a height of 2 cm and a width of 3cm.

$$\text{Volume} = 4 \text{ cm} \times 2 \text{ cm} \times 3 \text{ cm} = 24 \text{ cm}^3$$

$$\begin{aligned} \text{Density} &= \text{mass} / \text{volume} = 36 \text{ g} / 24 \text{ cm}^3 \\ &= 1.5 \text{ g} / \text{cm}^3 \end{aligned}$$

Apparatus:

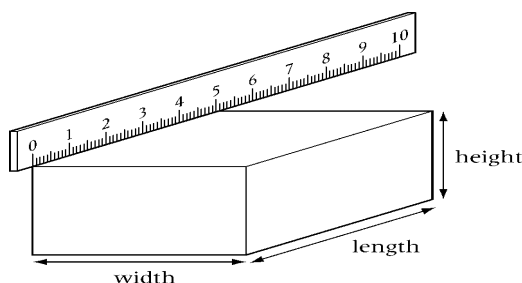
- Different materials that regular and irregular shapes.
- Bowl, displacement can and measuring cylinder.
- Ruler, Calculator & Electronic Balance

Method:

1. Make a table for your results, like this:

Material	Mass (g)	Length (cm)	Height (cm)	Width (cm)	Volume (cm^3)	Density (g/cm^3)	Floats or Sinks?

2. Choose a material, and write its name in your table.
3. Use a balance to find its mass, and write it in your table.
4. If the object is a regular cuboid, measure the length, height and width of your material, like this:



Write the measurements in your table.

5. Multiply the three measurements to find the volume of your material.

6. Divide the mass by the volume you have just worked out. This is the density. Write it in the correct column.
7. If the material has an irregular shape, your teacher will show you how to use the displacement can and a small measuring cylinder to find the volume.
8. Now see if your material floats or sinks. Write F or S in the last column.
9. Repeat steps 2 to 8 for other materials.

Conclusion:

What did you find out? Do materials with high densities float or sink?

Materials with a density greater than the density of water (1 g/cm³) sink, while materials with a density less than 1 g/cm³ float.

For example, iron (density 7.9 g/cm³) sinks, while wood (density 0.6 g/cm³) floats.

So high-density materials sink and low-density materials float.

Complete these sentences. Choose from the words in the brackets.

Density is a way of saying how much mass (mass/volume) there is in 1 cm³ of a material.

A material with a high density feels heavier (lighter/heavier) than a material with a low density.

Materials with a high density sink (float/sink) when you put them in water.

Materials with a low (high/low) density float. The density of water is 1 g/cm³.

· Diagram: show the large hollow volume of the ship displacing a large amount of water, producing a large upthrust.

If a material has a density less (less/greater) than the density of water, it will float.

Extension Questions: (Complete these on some paper and add them to your notes)

1. What is upthrust? Upthrust is the upward force that a liquid (or gas) exerts on an object placed in it.
2. If an object floats what is the relationship between the force of upthrust and the weight (the force due to gravity). Use a diagram to help explain.
3. If an object sinks what is the relationship between upthrust and weight? Draw another diagram to help explain.
4. If ships are often made of a metal, such as iron, which has a density far greater than water, why do they float? Draw a diagram to help explain.

When an object floats, the upthrust is equal to the weight - the two forces are equal and opposite, so the object stays still.

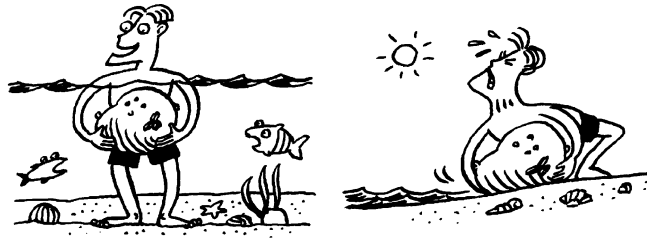
· Diagram: floating object with equal-sized arrows - upthrust (up) and weight (down).

When an object sinks, the weight is greater than the upthrust, so the object is pulled downwards.

· Diagram: sinking object with the downward weight arrow longer than the upward upthrust arrow.

7P 7b Floating questions

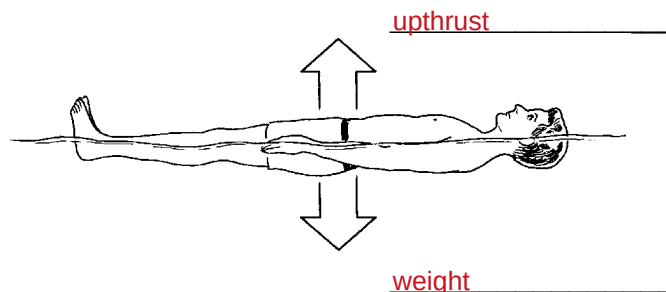
- 1 Josh was in the sea at high tide when he came across a large stone under the water. He tried to lift the stone and was surprised to find that he could move it easily. He returned at low tide and found the stone out of the water. This time he could hardly move it.



Use the words in the box to complete these sentences. Some words can be used more than once.

gravity	upthrust	water
	weight	

- a The stone felt lighter when it was under water. This is because upthrust was helping to push it up.
- b The stone does not float because its weight is greater than the upthrust.
- c The two forces which affect the stone are weight and upthrust.
- 2 This is Danny floating in the water. Label the two forces.



If Danny weighs 850N, what is the upthrust if he floats? 850 N.

3 Your weight is 600 N and you have just finished building a raft which weighs 1000 N.

a What is the total weight of you and the raft? 1600 N (600 N + 1000 N)

b The upthrust from the water is 1200 N. Will the raft sink or float? _____
It will float, because the upthrust (1200 N) is greater than the weight of the raft (1000 N).

c Will the raft sink or float when you get on it? sink Why?

Because the total weight of you and the raft is 1600 N, which is greater than the upthrust of 1200 N.

When the weight is greater than the upthrust, the object sinks.

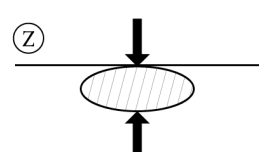
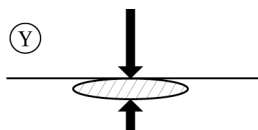
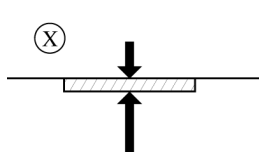
The raft will only float if the upthrust is greater than or equal to the total weight.

Extension: Is there a way to change the design of the raft, without changing the mass, which will help it float better?

Yes - make the raft wider and flatter so it displaces more water, which increases the upthrust without changing the mass.

7P 7c Floating (Extension)

- 1 What force holds a hot air balloon in the air?
- 2 Your weight is 600 N and you have just finished building a raft which weighs 1000 N.
 - a What is the total weight of you and the raft? **1600 N**
 - b What is the smallest upthrust force needed so that it will just float with you on it.
1600 N - the upthrust must be equal to the total weight for the raft to just float.
- 3 The pictures below show objects that have just been placed below the surface of the water. The arrows show the direction and size of the forces affecting each object.



- a Which object will sink? **The object where the downward weight arrow is larger than the upward upthrust arrow.**
 - b Which object will just float in the water?
 - c Which object will float well and be able to carry a small load?
The object where the upthrust arrow and the weight arrow are the same size (upthrust = weight).
- 4 The table shows the masses and volumes of pieces of different materials. Copy the table, and complete the last column. Remember the formula for calculating density is:

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

Material	Mass (g)	Volume (cm ³)	Density
9 g/cm ³ Copper	1800	200	
0.6 g/cm ³ Wood	900	1500	
11.34 g/cm ³ Lead	567	50	
7.9 g/cm ³ Iron	790	100	
1 g/cm ³ Water	1000	1000	
Polythene 0.92 g/cm ³	690	750	
2.46 g/cm ³ Glass	197	80	
Plasticine 1.3 g/cm ³	26	20	
Bone china 2.8 g/cm ³	42	15	
1.6 g/cm ³ Brick	3200	2000	

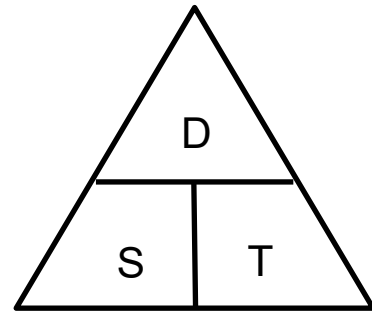
- 5 Make a list of all the materials in the table that will float.
The object where the upward upthrust arrow is larger than the downward weight arrow, so there is spare upthrust to carry a load.
Wood and polythene - their densities (0.6 g/cm³ and 0.92 g/cm³) are less than the density of water (1 g/cm³), so they float.

7P 8a Speed

Introduction:

- Speed is simply a measure of how quickly distance is covered.
- There are many different units for speed. The most common ones are metres per second (m/s), kilometres per hour (km/h) and miles per hour (mph).
- It can be calculated using the following equation:

$$speed = \frac{distance}{time}$$



Examples:

1. A car travels a distance of 2 000 metres in a time of 160 seconds. Calculate the average speed of the car in metres per second.

$$speed = \frac{distance}{time}$$

$$speed = \frac{2000}{160}$$

$$speed = 12.5 \text{ m/s}$$

2. How far will a cyclist travel in 60 seconds if he is travelling at an average speed of 13 metres per second?

$$distance = speed \times time$$

$$distance = 13 \times 60$$

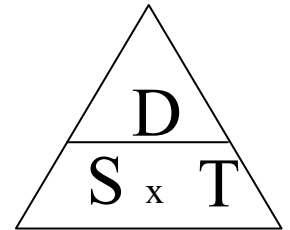
$$distance = 780 \text{ m}$$

Average speed = distance / time = 2000 m / 160 s = 12.5 m/s.

Distance = speed x time = 13 m/s x 60 s = 780 m.

7P 8b Calculating Speed

Speed can be calculated from the formula speed = distance/time.
The magic triangle can be remembered as “demented Science Teacher”



Units

If the distance is measured in metres and the time in seconds, speed will be calculated in metres per second (m/s).

If the distance is measured in kilometers and the time in hours, the speed will be calculated in kilometers per hour (km/h).

You may sometimes have to change units (for example, if you want a speed in kilometers per hour you may have to change 30 minutes into half an hour (0.5).

Use the above formula to answer the following questions. (don't forget to include units)

Easy questions

1. Mr Robertson's Ferrari (ROAR) goes a distance of 350 km in a time of 2 hours. What was his average speed in km/h?



2. Mr Windle and Kirsten Dunst are walking down the road. They walk a distance of 250 metres in 30 seconds. What was their average speed in m/s?

Medium questions

3. Mr Bayne, Mr Windle and Mr Robertson have robbed a bank. Their getaway car (the Ferrari) drives a distance of 45 kilometres at a speed of 15 km/h. How long did their journey take?



4. Jonny and James are going to a party together. They walk at a speed of 3.4 m/s for 63 seconds. How far away is the party? (in metres)

Hard questions

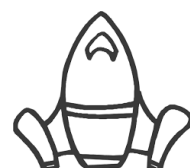
5. Jonathan and Madeleine have got married. On their honeymoon they drive 30 km in 15 minutes. What was their average speed in km/h?



6. Derek, Malene and Janne are walking to 7-11 to buy some soft drinks. They walk 3000 metres in 900 seconds. What was their average speed in km/h?

Very difficult question

7. A space ship travels 234500000 kilometres to planet Marit in 15 hours. Captain Angus wants to know the average speed of the space ship in m/s? What can you say about this speed?



$$\text{speed} = \text{distance} / \text{time} = 350 \text{ km} / 2 \text{ h} = 175 \text{ km/h}$$

$$\text{speed} = 250 \text{ m} / 30 \text{ s} = 8.33 \text{ m/s}$$

$$\text{time} = \text{distance} / \text{speed} = 45 \text{ km} / 15 \text{ km/h} = 3 \text{ hours}$$

$$\text{distance} = \text{speed} \times \text{time} = 3.4 \text{ m/s} \times 63 \text{ s} = 214.2 \text{ m}$$

7P 8c Measuring Average Speed Demonstration

Your teacher will demonstrate how to measure the average speed of a toy car down a slope.



Record the measurements and calculate the speed in the table below:

Distance (m)	Time 1 (s)	Time 2 (s)	Time 3 (s)	Average Time (S)	Average Speed (m/s)
	e.g. 1.00 m	e.g. 0.90 s	e.g. 0.85 s	e.g. 0.95 s	Average time = distance / average time, e.g. 1.00 m / 0.90 s = 1.11 m/s

Questions:

Answer the following questions about the demonstration.

- To calculate average speed, the distance and the time must be measured.

- What pieces of equipment were used to take these measurements?

A metre ruler (or tape measure) was used to measure the distance

A stopwatch was used to measure the time.

_____ The ruler/tape measure and the stopwatch are the pieces of equipment used.

- How was the reliability of the measurement improved?

The time was measured three times and the average was calculated.

Repeating the measurement reduces the effect of timing errors (e.g. reaction time)

Taking the average time gives a more reliable value to use in the speed calculation

7P 8d Practical Activities

You will now complete one or more of the following activities.

- Measuring the average speed of pupil moving in the playground.
- Measuring the average speed of a car as it travels along the road.
- Using a racing car game and measuring the average speed over one lap.

In the space on the next page, outline what you did and demonstrate the calculations that you used. Here are some possible things to include:

- A diagram of the set-up and apparatus.
- A description of the measurements that were taken.
- The equipment that was used to take the measurements.
- Any equations that were used to calculate average speed.

Extension:

- What the difference is between average and instantaneous speed? What is the difference in how they are measured?

Average speed = total distance / total time (whole journey); instantaneous speed is the speed at one moment (speedometer). One is measured over the whole journey, the other at a single instant.


MYP 1: Year 7 Science
Criterion B Assessment – Inquiring and Designing
Criterion C Assessment – Processing and Evaluating

UNIT:	Forces and Motion	
ASSESSMENT TITLE:	Investigating factors which affect the time taken for the helicopter to fall .	
Statement of Inquiry:	By exploring the laws of motion and relationships we can describe the movement and interactions of objects.	
Key Concept:	Relationships	
Related Concepts:	Movement and Interactions	
ATL Focus:	Information Literacy Skills Collaboration Skills	
Key Scientific Vocabulary:	Data	Measurement of a parameter that can be quantitative (volume, temperature, pH and so on) or qualitative (colour, shape, texture and so on).
	Dependent variable	The variable in which values are measured in the experiment.
	Prediction	Give an expected result of an upcoming action or event.
	Hypothesis	A tentative explanation for an observation or phenomenon that requires experimental confirmation; can take the form of a question or a statement.
	Independent variable	The variable that is selected and manipulated by the investigator in an experiment.
	Qualitative data	Refers to non-numerical data or information that is difficult to measure in a numerical way.
	Quantitative data	Refers to numerical measurements of the variables associated with the investigation.
	Validity of the method	Refers to whether the method allows for the collection of sufficient valid data to answer the question. This includes factors such as whether the measuring instrument measures what it is supposed to measure, the conditions of the experiment and the manipulation of variables (fair testing).
	Command Terms:	Design
Evaluate		Make an appraisal by weighing up the strengths and limitations.
Interpret		Use knowledge and understanding to recognize trends and draw conclusions from given information
Analyse		Break down in order to bring out the essential elements or structure. To identify parts and relationships, and to interpret information to reach conclusions.

Assignment Instructions (GRASPS)

Goal	You are required to determine how a named factor affects the time taken for the helicopter to fall .
Role	You are an aeronautical engineer who works in the field of helicopter design. You have been asked to investigate how a named factor affects the ability of the helicopter to fly . Is there an optimum length of blade? Is there an optimum weight? Cost is also an important consideration.
Audience	You will complete a scientific report on the findings of your investigation and submit it to the Head Engineer (your teacher!) for inspection.
Situation	You will work collaboratively in groups of twos or threes. You are required to design an organized practical investigation that will assist you in addressing the goal above. You may refer to the practical work that has been completed already in the Forces and Motion unit. You must then perform the practical you have designed so that you can collect results that will help you determine the relationship between the named factor and the time taken to fall/average falling speed? You will be given the following equipment to conduct your investigation: helicopter template, scissors, paper clips, meter Stick, tape measure and a stopwatch. Dependent Variable: the time taken to fall a set height (average falling speed = height / time taken).
Product	A scientific report should be completed to include a design, analysis and evaluation section. This report will be an electronic document that will be submitted via Google classroom for marking and will communicate the science of your investigation. Your teacher will advise you of the due date for the report.
Success Criteria for Assessment	The final scientific report MUST include (see MYP Practical Report Checklist for Students in your booklet): - Title - Research Question - Background information - Hypothesis - Variables - Apparatus - Method - Safety considerations - Results table - Observations - Calculations - Graph - Conclusion - Strengths and Weaknesses - Improvements Prediction: there should be an optimum blade length and an optimum weight (mass) that make the helicopter fall most slowly and steadily.

Level Descriptors and Task Specific Clarifications for Criterion B:

Level	Level Descriptor	Task Specific Clarification
0	The student does not reach a standard described by any of the descriptors below.	i. No research question stated. ii. No prediction stated. iii. Variables not identified. iv. Method and safety precautions not stated.
1-2	The student is able to: i. select a problem or question to be tested by a scientific investigation ii. select a testable prediction iii. state a variable iv. design a method with limited success .	i. A research question is stated but not relevant to the investigation. ii. A prediction with limited reference to forces. iii. Independent, dependent and controlled variables identified but with inaccuracies. Method of data collection not indicated. iv. A method is presented but incomplete or not relevant to the investigation. Safety precautions not mentioned.
3-4	The student is able to: i. state a problem or question to be tested by a scientific investigation ii. state a testable prediction iii. state how to manipulate the variables, and state how data will be collected iv. design a safe method in which he or she selects materials and equipment .	i. A relevant research question is stated, but is not too clear or focused. ii. A prediction with some reference to the variables is made. iii. Independent, dependent and controlled variables identified but incomplete. Basic method of data collection is stated. iv. Method is presented in steps but does not clearly show how the experiment is conducted and/or how data is collected. Safety precautions stated but are not specific to the experiment.
5-6	The student is able to: i. state a problem or question to be tested by a scientific investigation ii. outline a testable prediction. iii. outline how to manipulate the variables, and state how relevant data will be collected iv. design a complete and safe method in which he or she selects appropriate materials and equipment .	i. A research question is stated with reference to some correct variables. ii. A prediction with reference to the relationship between the independent and dependent variables is outlined (ie. the relationship between the named factor and the time taken for the helicopter to fall). iii. Independent, dependent and controlled variables correctly identified, with an outline of how relevant data is collected. iv. Method is presented in clear steps with relevant details, with suitable materials and apparatus chosen. A labelled diagram to show how the apparatus is set up. Relevant safety precautions discussed.

7-8	The student is able to: i. outline a problem or question to be tested by a scientific investigation ii. outline a testable hypothesis using scientific reasoning iii. outline how to manipulate the variables, and outline how sufficient, relevant data will be collected iv. Design a logical, complete and safe method in which he or she selects appropriate materials and equipment	i. A research question clearly and explicitly identifies the independent and dependent variables with reference ii. A prediction with reference to the relationship between the independent and dependent variables is outlined and explained using relevant scientific reasoning. The explanation should mention the size and direction of specific forces. iii. Independent, dependent and controlled variables correctly identified, outlining how they are manipulated, measured or controlled. Values of variables stated where appropriate. Outline how sufficient, relevant data is collected. iv. A complete method is presented in clear steps with all relevant details, stating values and measuring equipment used. A fully labelled diagram to show how the apparatus is set up. Safety precautions discussed and are relevant to this investigation.
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Level Descriptors and Task Specific Clarifications for Criterion C:

Level	Level Descriptor	Task Specific Clarification
0	The student does not reach a standard described by any of the descriptors below.	i. No data presented. No graph plotted. ii. Trend not stated. iii. No conclusion on whether the prediction is supported by data. iv. No strengths or limitations relevant to the investigation stated. v. No improvement or extension stated.
1-2	The student is able to: i. collect and present data in numerical and/or visual forms ii. interpret data iii. state the validity of a prediction based on the outcome of a scientific investigation with limited success . iv. state the validity of the method based on the outcome of a scientific investigation with limited success . v. state improvements or extensions to the method that would benefit the scientific investigation, with limited success .	i. Raw, quantitative data is presented in a table. A graph is attempted. ii. A trend is stated but is not supported by data. No scientific explanation given to explain the trend. iii. A conclusion is attempted , but with inaccuracies. iv. At least one limitation relating to the investigation is stated, but not relevant to the investigation. v. At least one improvement or extension stated but is not relevant to the limitation identified.
3-4	The student is able to: i. correctly collect and present data in numerical and/or visual forms ii. accurately interpret data and outline results iii. state the validity of a prediction based on the outcome of a scientific investigation iv. state the validity of the method based on the outcome of a scientific investigation v. state improvements or extensions to the method that would benefit the scientific investigation.	i. Raw, quantitative data is neatly presented in a table with headings. A graph is plotted with labelled axes. ii. Trend shown in the graph is outlined by stating a basic relationship between variables. iii. A correct conclusion is drawn with reference to the prediction. iv. At least one limitation relating to the validity of the investigation is stated. v. At least one improvement relating to each limitation raised is stated.
5-6	The student is able to: i. correctly collect, organize and present data in numerical and/or visual forms ii. accurately interpret data and outline results using scientific reasoning iii. outline the validity of a prediction based on the outcome of a scientific investigation iv. outline the validity of the method based on the outcome of a scientific investigation v. outline improvements or extensions to the method that would benefit the scientific investigation.	i. Raw, quantitative data from multiple trials are neatly presented in a table with headings and relevant units. An attempt has been made to calculate averages. A correct type of graph is plotted. The graph has correctly labelled axes including units and suitable scales. Some observations are recorded. ii. Trend shown in the graph is correctly outlined and explained using some correct scientific knowledge relating to forces. Anomalies/outliers are identified.

		iii. A conclusion is drawn. Student states if the data supports the prediction. iv. At least two limitations relating to the validity of the investigation are outlined. v. At least one improvement relating to each limitation is outlined. Extensions of investigation outlined.
7-8	The student is able to: i. correctly collect, organize, transform and present data in numerical and/ or visual forms ii. accurately interpret data and outline results using correct scientific reasoning iii. discuss the validity of a prediction based on the outcome of a scientific investigation iv. discuss the validity of the method based on the outcome of a scientific investigation v. describe improvements or extensions to the method that would benefit the scientific investigation.	i. Raw, quantitative data from multiple trials are neatly presented in a table with informative headings and relevant units. Averages are correctly calculated and presented in the table. A correct type of graph is accurately plotted. The graph has correctly labelled axes including units and suitable scales. A line of best fit is included. Meaningful observations are recorded. ii. Trend shown in the graph relating to the factor investigated is correctly outlined with reference to specific data points. Trend is explained using correct scientific knowledge relating to forces. Anomalies/outliers are correctly identified. iii. Correct conclusion is drawn as to how well the data supports the prediction, with reliability of the data discussed. iv. One strength and three limitations relating to the validity of the investigation are discussed. Consider how the limitations affected the data and how well the research question could be answered. v. At least one improvement relating to each limitation raised is described. Relevant extensions of investigation described.

Student Task Planning Notes:

Plan: investigate how blade length affects fall time. IV = blade length, DV = fall time, controls = mass, drop height and release method. Repeat each test 3 times and average.



MYP Science Practical Report Template

Name:

Form:

UNIT: Forces and Motion	
Statement of Inquiry:	
Key Concept: By exploring the laws of motion and relationships we can describe the movement and interactions of objects.	
Related Concepts:	
Investigation Title:	Investigating how the mass of a paper helicopter affects the time taken for it to fall.

Criterion B - Inquiring and Designing

Research Question How does the mass of the helicopter (number of paper clips) affect the time taken for it to fall?

(What are you trying to find out?)

Background Scientific Information and Hypothesis

(What do you think will happen and why?)

Gravity pulls the helicopter down; air resistance pushes up.

Hypothesis: the more paper clips, the shorter the fall time, because a larger weight overcomes air resistance more easily.

Variables

Independent <i>(What will change and how will you change it?)</i>	
Dependent <i>(What will you measure to gather result?)</i>	
Control <i>(What variables will you keep the same to make it a fair test? How will you keep the variables constant?)</i>	

The mass of the helicopter (number of paper clips). Changed by adding 1, 2 or 3 paper clips.

The time taken for the helicopter to fall to the ground, measured with a stopwatch.

Drop height, number of paper clips, type of paper and how the helicopter is released. Kept the same by using the same template, height

Apparatus Set-up

(Draw a labelled diagram of the assembled apparatus that you will use. Write the name of any equipment not shown in the diagram)

- Draw a labelled diagram of the apparatus: the paper helicopter with paper clips and the drop height; label the stopwatch.

Method

(Write a step by step account of how you conduct your experiment – the method must be DETAILED)

- 1.
- 2.
- 3.
- 4.
- 5.

1. Fold the template into a helicopter. 2. Attach one paper clip. 3. Drop from a set height and time the fall. 4. Repeat three times. 5. Repeat with two and three clips, keeping height the same.

Work in a clear space so nobody is hit by the helicopter; drop it from a safe height; cut out the template carefully with scissors.

Safety

(How will you make sure your method is safe?)

Reliability

(What will you do to ensure your results are reliable?)

Repeat each test three times and average the times; keep the drop height and release method the same; use the same stopwatch and timer each time.

Criterion C - Processing and Evaluating Results

Raw Data

Draw a results table with headings: number of paper clips, times for trials 1-3 (s) and average time (s); record all measurements with units.

(A neat, organized table of results should be drawn, with headings, units, repeats and averages include)

Observations

The helicopter spins as it falls; with more paper clips it falls faster and spins more quickly; it falls straight down without wobbling

(What changes did you notice as the experiment was conducted, e.g. colour change, bubbles, heat change?)

Graphing Results

You should draw a graph of your results and attach it to this report.

Analysis of Graph

The graph shows that as the number of paper clips (mass) increases, the time taken to fall decreases - a negative trend.

(What did your results show? Are there any trends or patterns in your results? Can you give an example of a pattern or trend from your results?)

Yes - as the independent variable increases the dependent variable changes steadily, so the points form a clear pattern/trend.

Conclusion and Evaluation

Scientific Conclusion and Explanation of Results

(Summarize what your results show. Does this support your prediction? Can you explain your results using scientific knowledge?)

The results support the prediction: more mass gives a shorter fall time, because a larger weight overcomes air resistance more easily, so the helicopter falls faster.

Evaluation of Scientific Method and Results

How reliable were your results? How well did your investigation answer the research question? What went well with your method?)

Results were reliable because tests were repeated and averaged. The investigation answered the research question (more mass = shorter fall time). The method was simple and easy to repeat.

(Were there any problems or parts of your investigation that were not very accurate or could have been done better?)

Issue with justification	Suggested SCIENTIFIC Improvement
Reaction time in starting/stopping the stopwatch made the times inaccurate. Improvement: use a light gate or video to time the fall automatically.	
Only one drop height was used, so the results may not apply to other heights. Improvement: repeat the investigation at several drop heights.	
The helicopter may have been released slightly differently each time. Improvement: always drop it the same way, flat and without pushing it.	
The blades may have been folded differently each time. Improvement: use the same template and fold all the helicopters identically with a ruler.	

7P 11a Investigating Distance-Time Graphs

Aim:

Constructing and interpreting simple distance/time graphs.

Apparatus:

- Stop watches
- Measuring tape / trundle wheel.

Method:

1. Measure out a course on the playground with a start position and then every 5 metres position someone with a stop watch. Ideally this should be about 40 or 50 metres long.
2. When the 'starter' shouts 'go', everybody starts their stop watch.
3. The first volunteer **walks** from the starting line to the finish line. When they pass the person with the stop watch, they stop the stop watch and record the time taken.
4. This is repeated but this time the second volunteer **jogs** slowly.
5. Finally, if you have time, the third volunteer can **run** quickly.
6. Record your results as a class.

Results:

Distance (m)	0	5	10	15	20	25	30	35	40	45	50
Walk (s)	0						20 s				36 s
Jog (s)	0						12 s				22 s
Run (s)	0						8 s				14 s

Analysis/Conclusion:

1. Draw a graph of distance (y axis/up) against time (x axis/across).
2. Plot on the same axis for the walk, jog and run graphs. Use different colours or label these.
3. What do you notice about the shapes of these graphs? Are they straight or curved? What does this mean?

All three graphs are straight lines, not curved.

Each line is straight because each volunteer moved at a steady speed.

The steeper the line the greater the speed - gradient = speed

4. What do you notice about the slopes of these graphs? Which was steepest? How does the steepness relate to the speed?

The faster the volunteer moved the steeper the straight line on the graph.

The run graph was the steepest because the runner had the greatest speed.

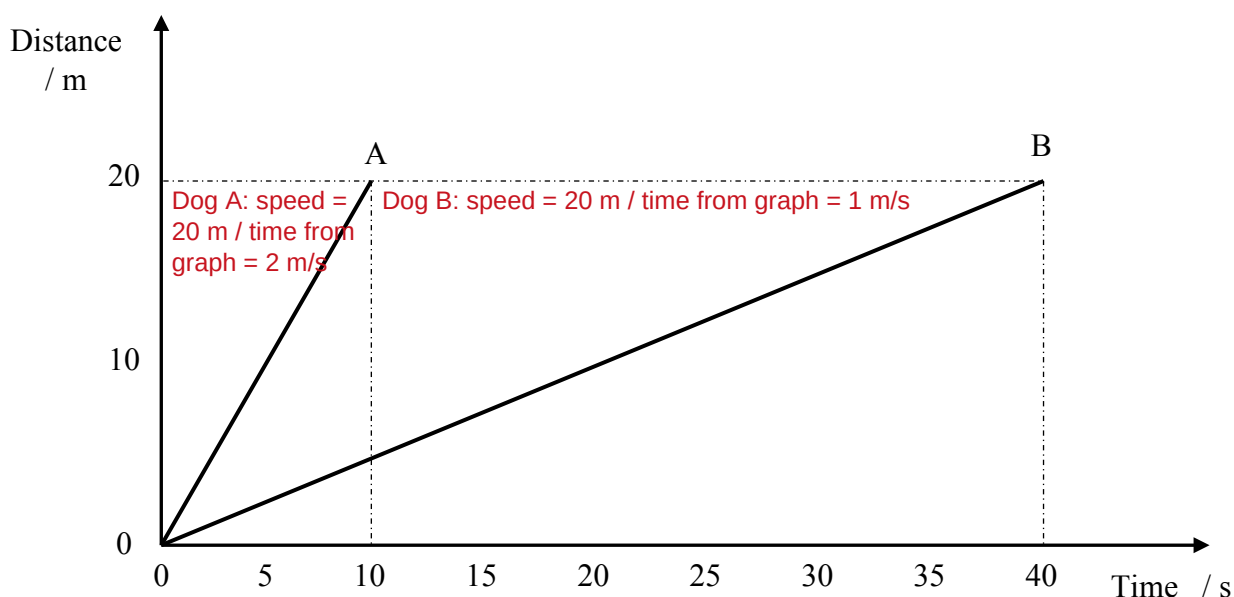
Speed = gradient of the graph, so a steeper slope means a higher speed

Extension:

Work out the speed of each of the volunteers from the graph. Can you do this from the gradient? Yes - speed = distance/time = the gradient. Walk about 1.2 m/s, jog about 3 m/s, run about 5 m/s (from your class results).

Practice Question:

Two dogs A and B walk 20m down the road.



Which dog is fastest and how can you tell from the graph?

Dog A is the fastest

Dog A's line reaches 20 m in a shorter time than dog B's line.

Dog A has the steeper line, and the gradient of the line is its speed

Calculate the speeds of both dogs.

Speed = distance / time. Read the time from the graph for each dog, e.g. A: $20 \text{ m} / 10 \text{ s} = 2 \text{ m/s}$; B: $20 \text{ m} / 20 \text{ s} = 1 \text{ m/s}$.

7P 12a – Moments

Introduction:

- A moment is a turning effect.
- A moment can also be called a TORQUE. (Torque is referred to when measuring car engines)
- The two things that affect the size of a moment are the sizes of the force and the perpendicular distance to the pivot.
- The pivot or axis of rotation is the point about which the object turns (eg the hinges of a door or the axle of a wheel)



The formula for moments is:

$$\text{moment} = \text{force} \times \text{perpendicular distance to the pivot}$$

The unit of moment is (Nm), but sometimes in the lab we may use (Ncm)

Examples of Moments:

- List some example of moments in everyday life:
Opening a door by its handle, turning a tap, using a spanner on a nut, a wheelbarrow, scissors, a seesaw, using a crowbar.

Example Calculation:

A door handle is opened by applying a force of 5N at a distance of 0.1m from the pivot. Calculate the Moment.

$$\text{moment} = \text{force} \times \text{perpendicular distance to the pivot}$$

$$\text{moment} = 5 \times 0.1$$

$$\text{moment} = \underline{0.5 \text{ Nm}}$$

7P 12b - Experimenting with Moments

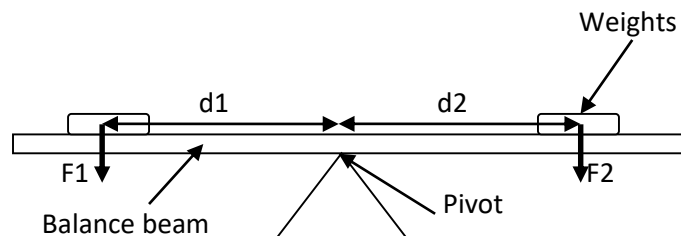
Part 1 – Establishing the Principle of Moments

Aim:

- By balancing different masses on a beam, on either side of the pivot, establish the principle of moments.

Apparatus:

- Balance beam
- 1N Weights
- Pivot



Method:

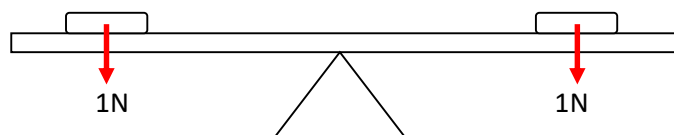
- For each combination of weights, try to balance them on either side of the pivot. Record the weight of each (see 'F1' and 'F2' on the above diagram).
- Measure the perpendicular distance from the pivot to the center of each weight (see labels 'd1' and 'd2' on the above diagram).
- Calculate the moment on both the left hand side and right hand side of the pivot using the following equation:

$$\text{moment} = \text{weight} \times \text{perpendicular distance from pivot}$$

- Compare the moment on the left hand side with the moment on the right hand side.

Weight Combinations:

- Balance 1N and 1N on the balance beam.



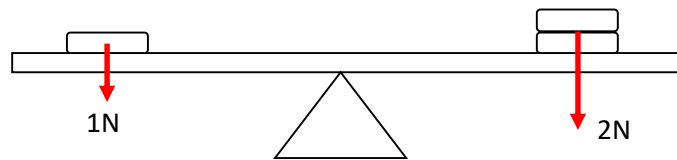
distance l.h.s = 20 cm

distance r.h.s = 20 cm

moment l.h.s = 1 N x 0.2 m = 0.2 Nm

moment r.h.s = 1 N x 0.2 m = 0.2 Nm

2. Balance 1N and 2N on the balance beam.



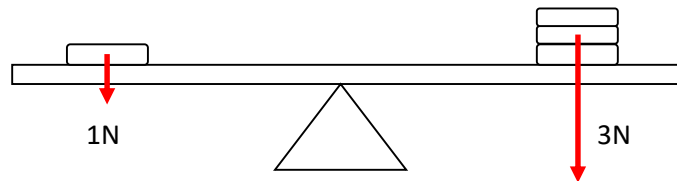
distance l.h.s = 40 cm

distance r.h.s = 20 cm

moment l.h.s = 1 N x 0.4 m = 0.4 Nm

moment r.h.s = 2 N x 0.2 m = 0.4 Nm

3. Balance 1N and 3N on the balance beam.



distance l.h.s = 60 cm

distance r.h.s = 20 cm

moment l.h.s = 1 N x 0.6 m = 0.6 Nm

moment r.h.s = 3 N x 0.2 m = 0.6 Nm

Conclusion

What do you notice about the moment on the left hand side and the moment on the right hand side? Can this be written mathematically?

The moment on the left hand side equalled the moment on the right hand side.

Yes: $F_1 \times d_1 = F_2 \times d_2$, so the anticlockwise moment equals the clockwise moment.

Evaluation:

Were there any problems in establishing your conclusion? What could be done to improve this experiment?

It was hard to judge exactly when the beam was level (balanced).

Repeat the readings and use small weights so the beam balances more easily.

Part 2 – Using the Principle of Moments to Make Predictions

Aim:

You will use the principle of moments to predict where to place weights to balance the beam.

Method:

- Your teacher will give you the weight and distance to the pivot for the weight on the left hand side. The weight on the right hand side will also be given.
- Use the principle of moments to calculate where to position the weight on the right hand side. In other words, its distance from the pivot.
- Place the object there to see if it balances. You may have to move it slightly.

Results:

- Use the space below to show your calculations and to record your results. Take note of the actual distance you had to place the weight on the right hand side for the beam to balance.

Left moment = weight x distance. Right distance = left moment / right weight (put in the numbers your teacher gives you).

Conclusion

1. Did the calculated distance exactly match the actual distance required to balance the beam? Yes/No. No - the actual distance was close to, but not exactly the same as, the calculated distance.
2. If not, can you explain why?

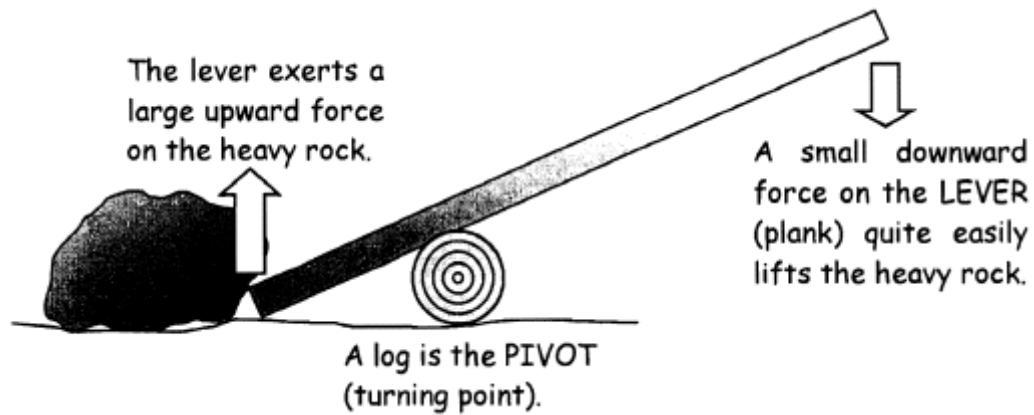
The beam itself has weight, which also has a turning effect about the pivot.

Friction at the pivot stops the beam turning freely.

The weight may not have been placed exactly at the measured distance (reading error).

7P 12c Levers

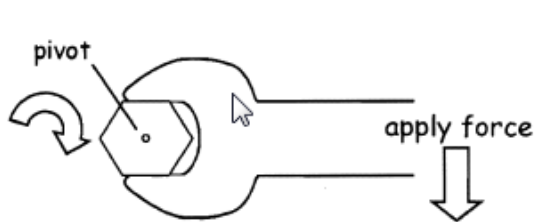
Forces can cause objects to turn around a pivot.



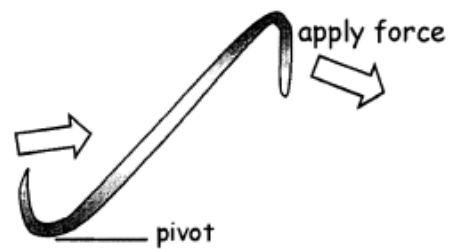
Important Rule:

The size of the turning force can be increased by increasing the length of the lever.

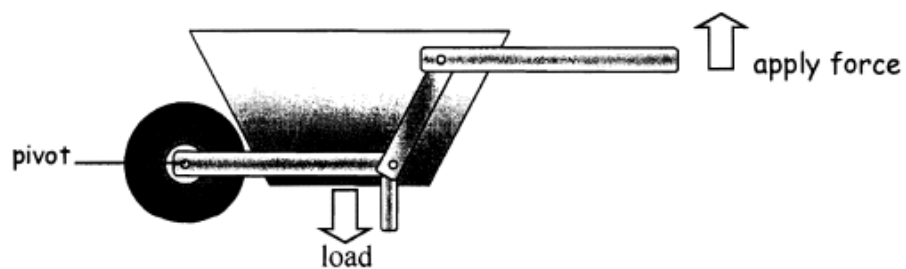
Examples:



Using a spanner to loosen a nut.



Using a crowbar to force objects apart.



Using a wheelbarrow to carry heavy loads.

· Examples of levers: crowbar, scissors, wheelbarrow, bottle opener, tweezers - each has a pivot, an effort and a load.

Demonstrations:

- You teacher will demonstrate some other levers in action. Draw a diagram of each demonstration and include an explanation of what is happening. Please use the space below:
 - Draw each lever in action and label the pivot, the effort and the load; explain that a long lever makes the turning effect bigger.

Questions:

1. A pivot is the turning point.
2. A longer lever makes it easier to move a heavy object.
3. The longer the lever the greater the moment it can exert about the pivot and also the greater the force it exerts on the load.
4. A crowbar can be used to lever open a locked door.
5. A tight nut can be loosened more easily if a longer spanner is used.

7P 13a Pressure

Introduction:

Pressure is the amount of force that is put onto a certain area.

$$\text{PRESSURE (N/m}^2\text{)} = \frac{\text{FORCE (N)}}{\text{AREA (m}^2\text{)}}$$

- The units for pressure are the newtons per square metre (N/m²)
- Another unit for pressure is the pascal (Pa)
- 1 N/m² = 1 Pa

Question:

Does the lady in stilettos or the elephant cause most damage to floor? Please explain your answer?

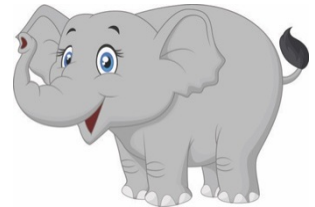


The lady in stilettos causes most damage to the floor.

Her weight is concentrated on the tiny area of the heel point.

A small area gives a very high pressure on the floor.

The elephant's weight is spread over a large area, so its pressure is lower.



Pressure can be made higher if the force is concentrated over a small area. A high pressure can cause an object to cut into a surface:



Pressure can be made lower if the force is spread over a large area. A low pressure will stop an object sinking into a surface.



Example Problems:

1. A table weighs 300N. The total area in contact with ground is 0.05m^2 . Calculate the pressure the table exerts on the floor.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

$$\text{pressure} = \frac{300}{0.05}$$

$$\text{pressure} = 6000 \text{ N/m}^2$$

2. An iceskater exerts a total pressure of $250,000 \text{ N/m}^2$ over an area of 0.002m^2 . Calculate the weight of the skater.

$$\text{force} = \text{pressure} \times \text{area}$$

$$\text{force} = 250000 \times 0.002$$

$$\text{force} = 500 \text{ N}$$

Questions:

Work out the answer to the questions below. Try to layout your answer as in the example problems above.

$$\text{Pressure} = \text{force} / \text{area} = 300 \text{ N} / 0.05 \text{ m}^2 = 6000 \text{ N/m}^2 \text{ (6000 Pa)}.$$

1. A man weighs 800N. The area of BOTH of his boots is 0.08 m^2 . What pressure does he place on the ground when he stands still?

$$\text{Weight} = \text{pressure} \times \text{area} = 250\,000 \text{ N/m}^2 \times 0.002 \text{ m}^2 = 500 \text{ N}.$$

$$\text{Pressure} = \text{force} / \text{area} = 800 \text{ N} / 0.08 \text{ m}^2 = 10\,000 \text{ N/m}^2 \text{ (10 000 Pa)}.$$

2. A woman weighs 500N. The area of ONE of her stiletto heels is 0.0002m^2 . What pressure does she place on the ground when she puts her weight onto one heel? $\text{Pressure} = \text{force} / \text{area} = 500 \text{ N} / 0.0002 \text{ m}^2 = 2\,500\,000 \text{ N/m}^2 \text{ (2 500 000 Pa)}.$

3. The base of a suitcase has an area of 0.2m^2 . It places a pressure of 700N/m^2 on the ground. What must the weight of the suitcase be?

$$\text{Force} = \text{pressure} \times \text{area} = 700 \text{ N/m}^2 \times 0.2 \text{ m}^2 = 140 \text{ N}.$$

7P 13b Calculating the Pressure you Exert on the Floor

Aim:

Calculate the pressure you exert on the floor by using the pressure equation. You will also need to use the 'counting the squares' method to find the area.



Apparatus:

- scales
- graph paper

Method:

1. Measure your weight using the scales. This should be recorded in Newtons.
 - (or by calculating weight = mass x g ; using g = 10 N/kg)
2. Students draw around one foot on graph paper and count 1cm squares to estimate area of foot.
3. The total area can be found by multiply the area of one foot by 2.
4. The total pressure the student is exerting on the floor can be calculated using:

$$pressure = \frac{force}{area}$$

Conclusion and Analysis:

1. What was the total pressure that you exerted on the ground?

Pressure = weight / total area of both feet, e.g. 500 N / 0.025 m² = 20 000 N/m²

Use your own measured weight and foot area - the value will vary.

2. How could you decrease the pressure you exert on the floor?

Spread your weight over a bigger area

For example sit or lie down, or wear flat shoes with a large area.

A larger area for the same weight gives a lower pressure.

Evaluation:

1. Suggest how you could improve the accuracy of your pressure calculation?

Draw around your foot more carefully and neatly on the graph paper.

Count the squares two or three times and take an average

Count part squares as half squares and whole squares as one.

Make sure your weight is measured in newtons (mass x 10).

Repeat the whole measurement and take a mean

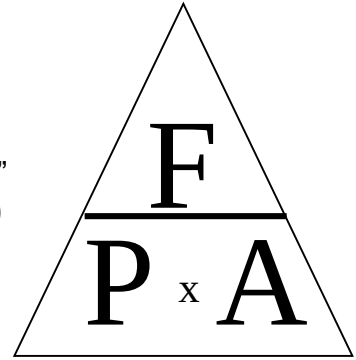
Use a finer grid so fewer squares are part-filled.

Remember to use the total area of BOTH feet.

7P 13c Calculating Pressure

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}}$$

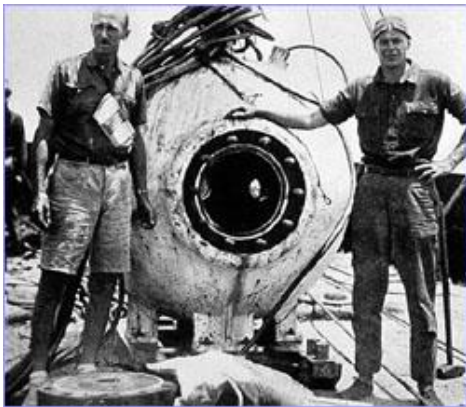
"football players argue"
(when under pressure)



The unit of pressure is N/m^2 (called a **Pascal**)

(Sometimes we use N/cm^2 as it can be more convenient)

Remember; Weight on earth (N) = mass (kg) x 10 (N/kg)



Did you know that the pressure at the deepest part of the Pacific ocean is about 110 million Pascals?!

The pressure of the atmosphere at sea level is approximately 101 000 Pa (it varies) and 100km in the air it is 1 Pa.



1. What is the pressure exerted by a student weighing 400N standing on one "stiletto" heel with an area of 1 cm^2 ? If an elephant weighs 9000N and each of its feet has an area of 200 cm^2 will it exert a greater pressure?

Student: $P = 400 \text{ N} / 1 \text{ cm}^2 = 400 \text{ N/cm}^2$. Elephant: $P = 9000 \text{ N} / 800 \text{ cm}^2$ (4 feet) = 11.25 N/cm^2 . No - the student's pressure is far greater.

2. If atmospheric pressure is $101\,000 \text{ N/m}^2$, what force is exerted on a door of area 2 m^2 ?

Force = pressure x area = $101\,000 \text{ N/m}^2 \times 2 \text{ m}^2 = 202\,000 \text{ N}$.

3. Explain why bulldozers use “caterpillar” tracks.

The caterpillar tracks have a very large area touching the ground.

The heavy weight of the bulldozer is spread over this large area.

So the pressure on the ground is low and it does not sink in.

The tracks also grip the ground so the bulldozer does not slip.



4. Explain why it hurts to hold a heavy parcel by one piece of string.

The string is thin, so it presses on a very small area of your hand.

A small area gives a very high pressure on your skin.

Use a wide flat handle or a glove to spread the force over a bigger area.

A bigger area gives a lower pressure, so it hurts less.



5. A rectangular block measures 8cm x 5 cm x 4cm and has a mass of 1.2 kg.

- a. What is the weight of the block? $\text{Weight} = \text{mass} \times 10 = 1.2 \text{ kg} \times 10 \text{ N/kg} = 12 \text{ N}.$
- b. What is the area of the smallest face of the block? $\text{Smallest face} = 5 \text{ cm} \times 4 \text{ cm} = 20 \text{ cm}^2.$
- c. What pressure will the block exert if it is resting on its smallest face?
 $\text{Pressure} = 12 \text{ N} / 20 \text{ cm}^2 = 0.6 \text{ N/cm}^2.$
- d. What is the least pressure the block would exert when resting on a table?
 $\text{Largest face} = 8 \text{ cm} \times 5 \text{ cm} = 40 \text{ cm}^2, \text{ so least pressure} = 12 \text{ N} / 40 \text{ cm}^2 = 0.3 \text{ N/cm}^2.$

6. A coin is being pushed into plasticine in two ways as shown:

- a. Explain why it is easier in diagram A to push the coin into the plasticine.

In diagram A the coin is pushed in on its edge, which has a very small area.

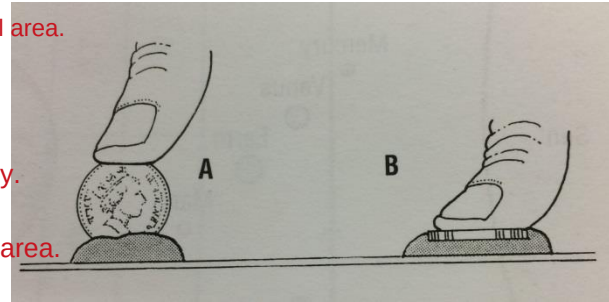
A small area gives a high pressure on the plasticine.

This high pressure makes the coin sink into the plasticine easily.

In diagram B the coin is pushed on its flat face, a much bigger area.

The bigger area gives a lower pressure, so it is harder to push in.

On its edge the coin has a much smaller area, so the same force gives a greater pressure and it sinks deeper into the plasticine.



- b. In diagram B, the area of the face of the coin is 2cm^2 . The finger is pushing with a force of 10N . What pressure is exerted on the plasticine? Give the unit.

Pressure = force / area = $10\text{ N} / 2\text{ cm}^2 = 5\text{ N/cm}^2$.

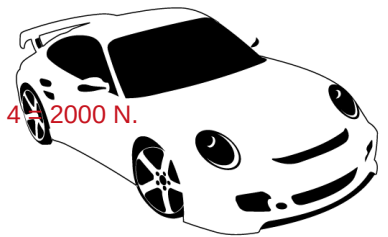
7. A man weighs 800N . Each foot has an area of 200cm^2 . What is the pressure exerted on the ground when he stands on:

- a. One foot? One foot: $P = 800\text{ N} / 200\text{ cm}^2 = 4\text{ N/cm}^2$.
- b. Both feet? Both feet: $P = 800\text{ N} / 400\text{ cm}^2 = 2\text{ N/cm}^2$ (the area is doubled so the pressure halves).



8. A car weighs 8000N . It has 4 wheels. Each tyre has a pressure of 20N/cm^2 .

- a. If the weight of the car is spread evenly, what is the weight supported by each tyre? Weight on each tyre = $8000\text{ N} / 4 = 2000\text{ N}$.



- b. Calculate the area of each tyre in contact with the road.

Area = force / pressure = $2000\text{ N} / 20\text{ N/cm}^2 = 100\text{ cm}^2$ per tyre.

- c. When some people get in the car, the area of contact for each tyre increases to 125cm^2 . If the pressure of each tyre is still 20N/cm^2 , what is the total weight of the car and passengers now? Force per tyre = $20\text{ N/cm}^2 \times 125\text{ cm}^2 = 2500\text{ N}$, so total weight = $2500\text{ N} \times 4 = 10\,000\text{ N}$.

9. A man is struggling to walk over soft snow. Explain why it is difficult to walk over soft snow.



His weight acts on the small area of his feet, so the pressure is high and his feet sink into the soft snow.

This lady has invested in a pair of snow shoes. Explain why it is easier for her to walk over soft snow.

Snow shoes have a large area, so the same weight gives a much lower pressure and she does not sink.



7P 14a Forces and Motion Quick Quiz

On your answer sheet, circle the correct letter for each question.

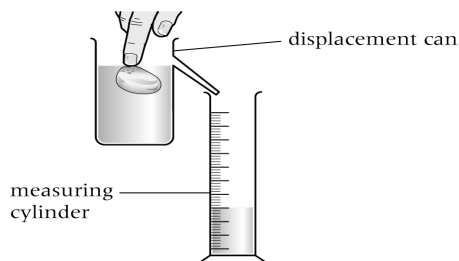
Section A

- 1 A force is:
A a spring. B an engine.
C a movement. D a push or a pull.
- 2 Which answer shows three non-contact forces?
A magnetism, gravity, friction
B gravity, friction, upthrust
C gravity, static electricity, magnetism
D upthrust, static electricity, friction
- 3 The unit of force is the:
A newton B kilogram.
C gram. D metre.
- 4 A force can be measured using:
A a stop clock. B a ruler.
C a thermometer. D a force meter

Section B

- 1 A boat will float because:
A gravity does not work over water.
B it is built out of light materials.
C upthrust pushes against its weight.
D it has air inside it.
- 2 Density is:
A the weight of a fixed mass of something.
B the volume of a fixed weight of something.
C the weight of a fixed volume of something.
D the mass of a fixed volume of something.

3



You can find the volume of the object in the can by:

- A measuring the object and multiplying the numbers.
- B measuring the volume of water it displaces.
- C weighing the object.
- D using a thermometer.

4 The density of water is 1 g/cm^3 . Which of these objects will float in water?

- A density = 2 g/cm^3
- B weight = 2 N
- C density = 0.5 g/cm^3
- D mass = 1 g

Objects with a density less than 1 g/cm^3 (e.g. wood, ice) float; objects with a density greater than 1 g/cm^3 (e.g. metal) sink.

Section C

1 An elastic material:

- A does not stretch.
- B stretches and stays in its new shape.
- C stretches and then goes back to its original shape.
- D cannot be squashed.

2 What are the units for weight and mass?

- A weight – newtons;
mass – kilograms
- B weight – kilograms;
mass – newtons
- C weight – kilograms;
mass – kilograms
- D weight – newtons;
mass – newtons

3 Which is true?

Weight is a force measured in newtons with a force meter (spring balance); mass is the amount of matter measured in kilograms.

- A Your weight would be the same on the Earth and on the Moon.
- B Your mass would be the same on the Earth and on the Moon.
- C Your weight would be more on the Moon than on the Earth.
- D Your mass would be less on the Moon than on the Earth.

4 What do most force meters have inside them?

- A a light bulb
- B a spring
- C wires
- D a battery

Section D

1 Which of the following sentences about friction is **not** correct?

Not correct: 'friction is always a nuisance and never useful' - friction gives grip (brakes, shoes) and also produces heat and wear.

- A Friction can make an object get hotter.
- B Friction wears car tyres away.
- C Friction slows down a ball rolling in grass.
- D Friction can speed up a falling object.

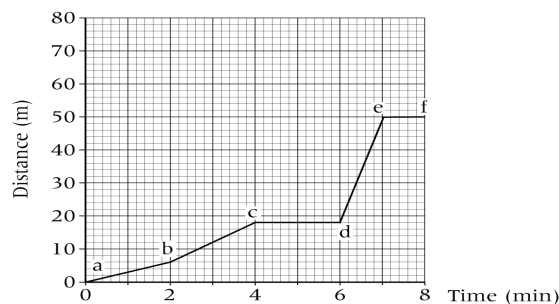
2 Which of these would produce the most friction when they rub together?

- A a rough surface and a smooth one
- B two smooth surfaces
- C two rough surfaces
- D two smooth surfaces with oil between them

- 3 Which example shows unhelpful friction?
- A friction between a tyre and the road
 - B friction in the wheel axles of a bicycle
 - C friction between paper and the point of a pencil
 - D friction between your shoes and the floor
- 4 Which of the following is **not** an example of a lubricant?
- A grease for bicycle chains
 - B rubber on car tyres
 - C water on swimming pool tiles
 - D oil in a car engine

Section E

- 1 The speed of a car is a measure of how fast it is travelling. Which is the missing word?
- A far
 - B fast
 - C long
 - D high
- 2 Air resistance increases if the surface area of an object increases.
- A mass.
 - B surface area
 - C density.
 - D cost.
- 3 What does 'streamlined' mean?
- A able to find the answer quicker
 - B has a shape that reduces air resistance and helps it travel faster
 - C has a really small mass some can move with a greater speed.
 - D is very intelligent and able to find their way to a location.
- 4 This graph shows the distance moved by someone catching a bus.



Which part of the graph shows the person waiting at the bus stop?

- A a to b
- B b to c
- C c to d
- D d to e

The one that does not reduce friction, e.g. sand - lubricants are oil, grease, wax and graphite, which form a smooth layer and lower friction.

Shaped so that air or water flows smoothly past the object, reducing air resistance (e.g. the curved, pointed shape of a fish or racing car).

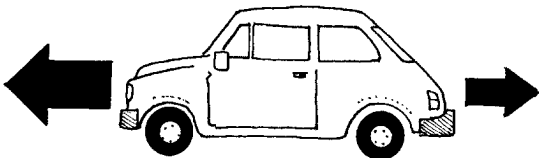
The flat (horizontal) part of the graph, where the distance stays the same while time passes, so the speed is zero.

Section F

- 1 The speed of a car is a measure of how:
- A far it is travelling.
 - B fast it is travelling.
 - C long it is travelling.
 - D high it is travelling.
- 2 A bus travelled a distance of 20 km in 2 hours. Its speed was:
- A 0.1 km/h.
 - B 40 km/h.
 - C 20 km/h.
 - D 10 km/h
- 3 The mean speed of something is:
- A its fastest speed.
 - B the total distance it travels divided by the time taken.
 - C its slowest speed.
 - D its speed limit.
- 4 If a car travels at 50 mph, how long will it take to travel 75 miles?
- A 1.5 hours
 - B 0.5 hours
 - C 0.67 hours
 - D 67 minutes

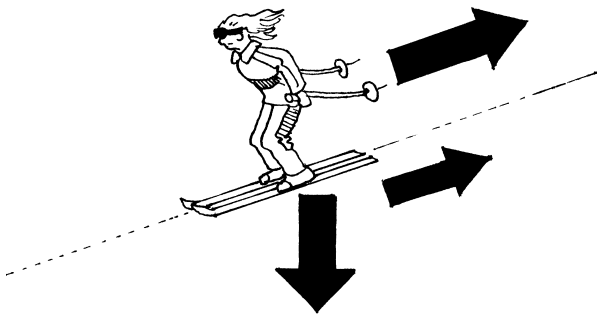
Section G

- 1 Balanced forces:
- A make objects speed up.
 - B make objects slow down.
 - C do not change the speed of moving objects.
 - D only affect stationary objects.
- 2 This drawing shows two forces on a car. What will happen to the speed of the car?
- The forces are unbalanced, so the car speeds up (accelerates) in the direction of the larger force.



- A It will accelerate.
- B It will slow down.
- C It will decelerate.
- D It will stay the same.

- 3 What are forces X and Y?



- A X is air resistance, Y is gravity.
B X is air resistance, Y is friction.
C X is gravity, Y is air resistance.
D X is friction, Y is air resistance.
- 4 An object will accelerate quickly if it has:
A a large force and a small mass.
B a large force and a large mass.
C a small force and a large mass.
D a small force and a small mass.

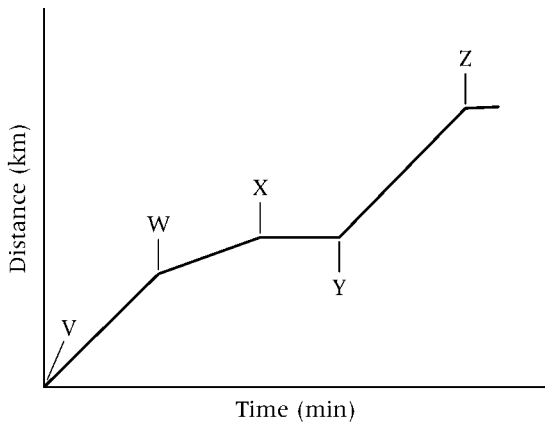
Section H

- 1 The air resistance of a car can be reduced by:
A making the car bigger.
B making the car heavier.
C giving it a streamlined shape.
D putting a roof rack on it.
- 2 Which statement about air resistance is true?
A If speed increases, air resistance decreases.
B If speed increases, air resistance increases.
C If speed decreases, air resistance increases.
D Speed has no effect on air resistance.
- 3 A car needs energy to keep moving because:
A everything needs a force to keep moving.
B gravity is trying to slow it down.
C it needs a force to balance friction and air resistance.
D weight balances friction.
- 4 Air resistance on a moving object is caused by:
A air being attracted to the object.
B air particles moving around the object.
C the wind.
D air particles hitting the object.

Section I

- 1 There are changing forces on a skydiver during a jump. Which of these statements is **not** true?
- A Her weight stays the same.
 - B Her air resistance stays the same.
 - C Her air resistance is very small at the beginning of the jump.
 - D Her air resistance increases as her speed increases.
- 2 A skydiver reaches a maximum speed when:
- A his air resistance has increased to balance his weight.
 - B his air resistance has decreased to balance his weight.
 - C his weight has increased to balance the air resistance.
 - D his parachute opens.
- 3 This distance–time graph shows Mrs Martin driving her children to the cinema. Which part of the graph shows where she stopped at some traffic lights?

The flat, horizontal part of the distance-time graph, where the distance does not change as time passes (speed = 0).



- A V to W
- B W to X
- C X to Y
- D Y to Z

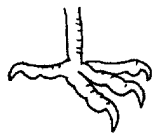
Section J

1 Why is it easier to cut something with a sharp knife than with a blunt one?

- A The area of the knife edge is large so pressure is greater.
- B The area of the knife edge is small so pressure is greater.
- C The area of the knife edge is large so pressure is smaller.
- D The area of the knife edge is small so pressure is smaller.

A sharp knife has a much smaller area at its edge, so the same force produces a much greater pressure and it cuts more easily.

2 If all these birds are the same weight, which bird has the lowest pressure under its feet?



A crow



B seagull



C raven



D chicken

3 Which is **not** a unit for pressure?

- A pascals
- B newtons per square metre
- C kilograms
- D newtons per square millimetre

4 A basketball player weighs 500 N. The area of one of his trainers is 250 cm². When he puts all his weight on one foot, how much pressure does he put on the ground?

- A 0.5 N/cm²
- B 2 N/cm²
- C 250 N/cm²
- D 125 000 N/cm²

The newton (N) - pressure is measured in pascals (Pa) or newtons per square metre (N/m²), not in newtons.

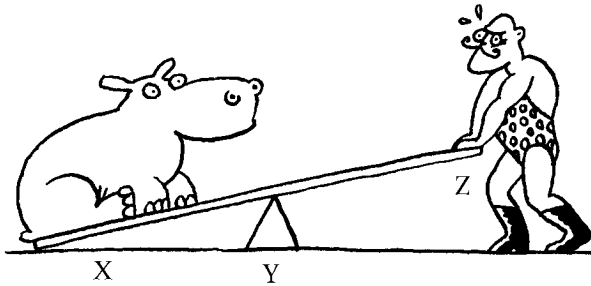
$P = F / A$: divide his weight in newtons by the area of one foot in m². It is twice the pressure of standing on two feet.

Section K

1 In science, a lever is:

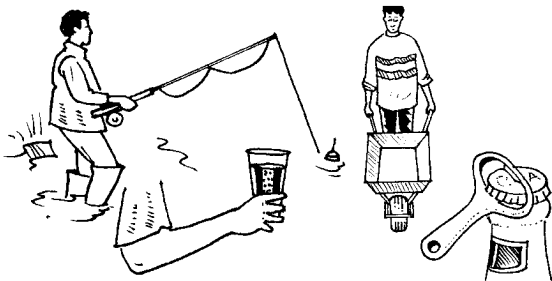
- A a complex machine which keeps the size of force you put in the same.
- B a simple machine which changes the size of force you put in.
- C a simple machine which changes the area of force you put in.
- D someone about to finish school.

2 Which are the correct labels for the lever shown below?



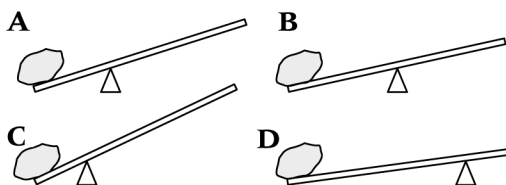
- A X = effort, Y = pivot, Z = load
- B X = pivot, Y = effort, Z = load
- C X = load, Y = pivot, Z = effort
- D X = load, Y = effort, Z = pivot

3 How many diagrams show levers in action?



- A 1
- B 2
- C 3
- D 4

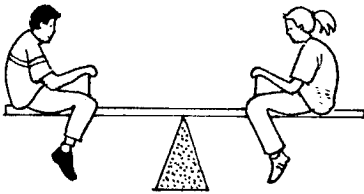
4 Which lever will make it easiest to move the rock?



· The lever with the longest arm between the effort and the pivot (pivot nearest the rock) is easiest - choose that diagram.

Section L

- 1 The moment of a force is:
- A its turning effect
 - B how long it takes to work.
 - C how far it moves.
 - D the distance of a force from a pivot.
- 2 You are opening a can of paint with a screwdriver. You apply a force of 10 N, 20 cm from the pivot. What is the moment of the force?
- A 2 Nm
 - B 20 Nm
 - C 200 Nm
 - D 2000 Nm
- 3 What is a seesaw that is 'in equilibrium'?
- A a balanced seesaw
 - B a seesaw with the same weight on each side
 - C a seesaw with both people the same distance from the middle
 - D a wooden seesaw
- 4 What is the distance from the pivot of the boy on the seesaw?



Boy: 300 N ☐ ? m Girl: 450 N ☐ 2 m

- A 1 m B 2 m C 3 m D 4 m

The perpendicular distance from the pivot to the boy, read off the diagram - it is the distance measured along the seesaw.

Section M

1 Acceleration is:

- A how much the speed changes each second
- B how long it takes an object to reach top speed
- C when something is staying the same speed
- D the distance it moves each second

2 The acceleration of an object depends on:

- A the size of the object
- B the force acting.
- C the speed of the object.
- D the unbalanced force and mass.

3 A force of 45N is exerted on an object of 9kg. What is the resultant acceleration?

- | | |
|-----------------------|-----------------------|
| A 405 m/s^2 | B 0.2 m/s^2 |
| C 5 m/s^2 | D 54 m/s^2 |

4 An object of mass 4kg accelerates at 2 m/s^2 . What is the size of the unbalanced force causing this acceleration.

- | | |
|--------|------|
| A 0.5N | B 8N |
| C 2N | D 6N |

7P Forces and Motion Summary Sheets

Forces and their effects

Forces are **pushes, pulls or twists**.

Forces can:

- change the **shape** or **size** of an object
- change the **speed** things are moving (make them move faster or slower)
- change the **direction** of a moving object.

The units for measuring force are **newtons (N)**.

Friction is a force caused by two things rubbing together. **Air resistance** and **water resistance** are kinds of friction. They are sometimes called **drag**.

Upthrust pushes things up. Solid things, like your chair, give you upthrust. Things float in water because of upthrust.

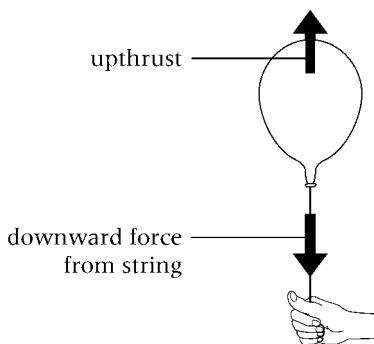
Contact forces need to touch the thing that they are affecting. Examples of contact forces are:

- friction
- air resistance
- water resistance
- upthrust.

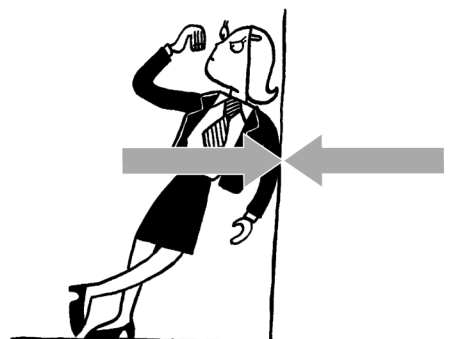
Some forces do not need to touch the thing that they are affecting. They are called **non-contact** forces. There are three non-contact forces:

- **magnetism**
- **gravity**
- **static electricity.**

Balanced forces



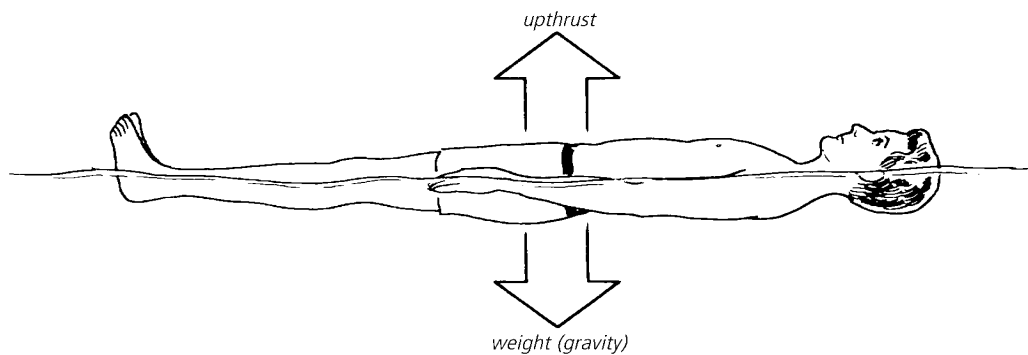
*The upwards and downwards forces on this balloon are **balanced**. The balloon will not move.*



The forces here are balanced. The girl will not move, and neither will the wall!

A rocket in space does not need to use its engine to keep moving. There is no air in space, so there is no air resistance to slow it down.

If you are floating in a swimming pool, your weight and the upthrust are balanced.



Density and floating

You can decide if something will float by working out its **density**. Density is the mass of a certain volume of something, and it can be calculated using this formula:

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

The units for density are g/cm^3 .

The density of water is 1g/cm^3 . If an object has a density less than 1g/cm^3 it will float. If its density is greater it will sink.

Newton's 1st Law

An object will continue to move in a **straight line** at a **constant speed** unless acted upon by an **unbalanced force**. In other words:

Balanced forces: constant speed

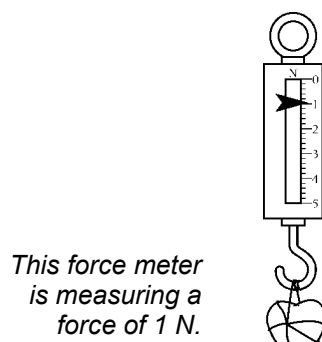
Unbalanced forces: acceleration.

Measuring forces

Elastic materials will stretch with a force and then return to their original shape when the force is taken away.

Materials like Plasticine will stretch with a force but they will not return to their original shape afterwards. Plasticine is not elastic.

Springs are used to measure the size of a force because they are elastic. A big force stretches a spring further than a small force. **Force meters** have springs inside them.



Weight and mass

Your **mass** is the amount of substance in your body. Your mass is measured in **kilograms (kg)**.

Your weight is a force caused by gravity pulling on your body. The **newton (N)** is the scientific unit used to measure forces, and so it is also used as the unit for weight.

Wherever you take an object, its mass will not change but its weight depends on the force of gravity. An object on the Moon would have a smaller weight than on Earth, because the Moon's gravity is not as strong as Earth's.

On Earth, gravity pulls on every kilogram of mass with a force of 10 N.

The equation is **weight = mass x 10**

Friction: Friction is a contact force. Friction can:

- slow things down
- wear things away
- produce heat
- make a noise.

Friction is sometimes helpful, for instance:

- your shoes grip the floor because of friction
- tyres and brakes use friction
- pencils write because of friction.

Friction is not always helpful:

- parts of engines wear away because of friction
- friction makes bicycles harder to pedal.

Friction can be increased by using rough surfaces, or by using materials like rubber that have a lot of friction.

Friction can be reduced by using smooth surfaces, or by **lubrication**. Things like oil or grease are **lubricants**, and help things to move past each other easily.

Static friction exists when two surfaces are in contact, but not sliding.

Sliding friction exists when two touching surfaces are sliding over one another.

Static friction is higher than sliding friction.

Static Electricity

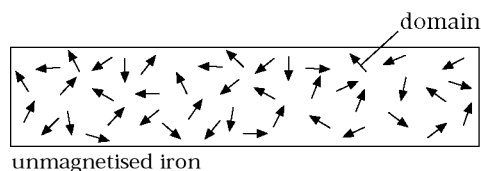
Static charge can build up when two surfaces rub against each other. **Electrons move from one object to another.** These extra electrons make the object negatively charged. The other object is now positively charged.

Object with the same type of charge will repel. Objects with opposite charge will attract each other.

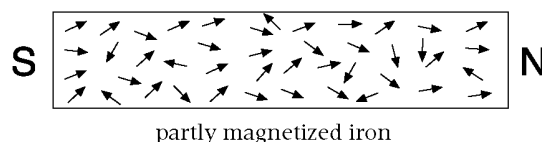
Magnetism

Magnetism is a property of the atoms in iron, nickel and cobalt. Little groups of atoms act like tiny magnets. Each little group of atoms is called a **domain**.

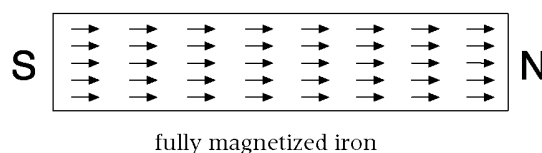
A normal piece of iron does not seem to be magnetic, because all the **domains** are **magnetised in different directions**. They cancel each other out.



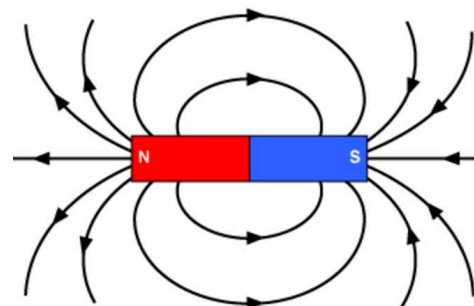
When a piece of iron is magnetised all the **domains line up**. They do not cancel each other out any more.



You can **magnetise** a piece of iron by **stroking** it. You always have to stroke in the same direction. You can also magnetise a piece of iron by leaving it next to another magnet.



Magnetic field lines show the **direction** and indicate the **strength** of the magnetic force. The **closer the lines** are together, the **stronger the force**.



Air Resistance

The larger the surface area, the larger the air resistance.

This is what happens when a sky diver, pull their parachute. The increase in air resistance is larger than the force due to gravity. This causes the sky diver to slow down, so they can land safely.

Certain cars, aero planes and many animals are adapted to allow them to reduce their air resistance by having a shape with a low surface area. We say that they have a **streamlined** shape.

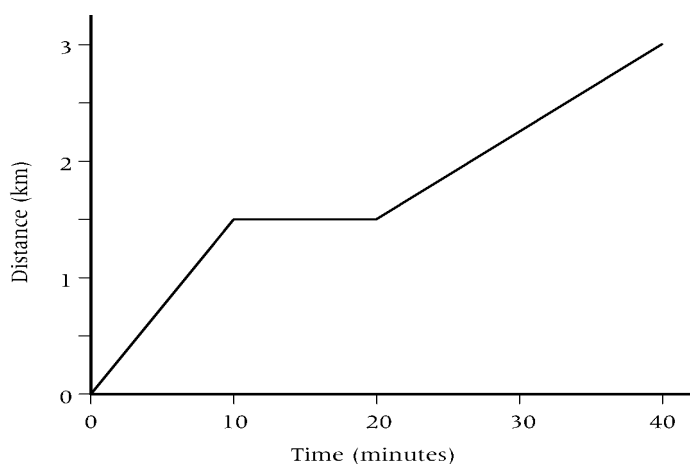
Speed

To measure how fast something is travelling you need to measure the distance it travels and the time taken. Units of speed are km/h or **m/s** or mph. The units for speed depend on the units you have used to measure the distance and the time.

Speed = distance (m) / time (s)

Distance/time graphs

A journey can be shown on a distance/time graph. This graph shows a person running, then stopping for a rest, then walking slowly. The **steeper the line on the graph, the faster they are moving**.



Moments

A moment is a **turning effect**. A moment can also be called a **TORQUE**.

The two things that affect the size of a **moment** are the sizes of the **force** and the **perpendicular distance** to the pivot.

The **pivot** or **axis of rotation** is the point about which the object turns (eg the hinges of a door or the axle of a wheel)

The formula for moments is:

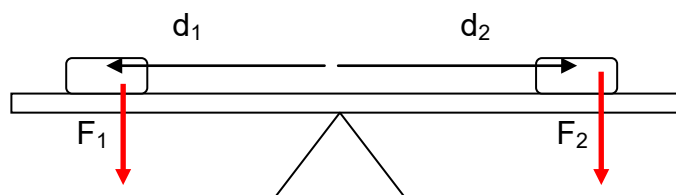
$$\text{moment} = \text{force} \times \text{perpendicular distance to the pivot}$$

The unit of moment is Nm.

Principle of Moments

If an object is balanced:

Moments left hand side of the pivot balance the moments right hand side of the pivot



As an equation this becomes:

$$F_1 \times d_1 = F_2 \times d_2$$

The size of the turning force can be increased by increasing the length of the lever.

Pressure

Pressure is the amount of force that is put onto a certain area. The equation is:

Pressure = force / area. The units are **N/m²** or **Pa**.

Pressure can be made **higher** if the force is concentrated over a **small area**:

Pressure can be made **lower** if the force is spread over a **large area**.

Newton's 2nd Law

An object will have a **greater acceleration** if there is a **larger unbalanced force** or a **smaller mass**. The equation is:

unbalanced force = mass x acceleration

The units for acceleration are **m/s²**.

7P 17a Forces Group Research and Presentation – Student Instruction Sheet

Aim:

You are required, in a group, to prepare a presentation on one of the forces being studied in the Forces and Motion unit. This presentation will be an assessed piece of scientific work.

Topics:

Your group may choose **one** from the following list of forces to research and present on;

1. Weight/Gravity
2. Friction
3. Magnetic Force
4. Buoyancy
5. Static Electricity
6. Turning Forces

Guidelines:

- Student groups will consist of **3, 4 or 5** members.
- You can choose any **ICT based method of presentation** that you are comfortable with (e.g. PowerPoint, web based, prezzi etc.)
- You should try and embed some pictures or video clips into your presentation that **reflect the application** of your chosen.
- You need to be prepared to present your project to the rest of the class. The maximum time for your presentation is **5 mins**.
- You will be assessed on your **teamwork/collaboration skills** as well as the quality of your presentation and how you **discuss the application of your chosen force to solve real life problems**.

Hints for Productive Teamwork:

1. Organise your team, make sure everyone knows what they need to do and when they need to be ready.
2. Elect a team leader who takes responsibility to ensure everyone on the group is involved in completion of the task
3. Discuss ideas, try to listen to other opinions. If you don't agree try to find some middle ground or a compromise when you make decisions.
4. Make sure you find a good way to communicate or share ideas. If someone is needs help, make sure they can get some support and encouragement.

Hints for a Good Quality Presentation:

1. **Research the force chosen thoroughly** by considering some of the following:
 - Describe the force. What causes it to happen? Give some examples of where/when we may have experienced it. How is it measured?
 - Identify 3 uses/applications of your chosen force? Explain the science behind each application.
 - For each use/application, describe the impact it has on the environment and humanity.
2. Make sure you **organise your presentation into relevant sections**. Try and use pictures/diagrams/short videos/links to help explain.
3. You need to make some **research notes by hand or on your computer**. It is really important that you do not cut and paste sections from the internet. You need to explain the concepts in your **own words**. Try to use **scientific language accurately**.
4. Ensure that you are using **multiple reliable sources**, and document the sources by including a **bibliography**.
5. When you have completed your presentation, try and **evaluate and reflect** on how well you and your team worked together.

Appendix - answers with no space on the original page

These belong to the printed question on the page shown.

p34: Clips attract strongly when unlike poles meet (A to B); when like poles meet (A to A) they repel or hardly attract at all.

p42: Upthrust - the upward force from the surrounding air. The hot air inside the balloon is less dense than the cold air outside, so the upthrust is greater than the weight.

p44: 15 minutes = 0.25 h; speed = 30 km / 0.25 h = 120 km/h

p44: speed = 3000 m / 900 s = 3.33 m/s = 12 km/h

p44: speed = 234 500 000 km / 15 h = 15 633 333 km/h = 4 342 593 m/s; this is extremely fast - much faster than any vehicle or aircraft on Earth.

p78: Two rough surfaces, e.g. rubber or rough wood rubbed together - rough surfaces grip more and so give the most friction.

p79: Friction between moving machine parts that causes heat and wear, so the machine needs oiling.

p81: Air resistance is a contact force that opposes motion, and it increases with the speed of the object and with its surface area.

p84: Count the diagrams where a force turns an object about a pivot (e.g. seesaw, crowbar, spanner, scissors, wheelbarrow).